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Internship report

Scattering of electromagnetic waves by the turbulent structures of a TORPEX plasma

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Abstract

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by John Smith

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Chapter 1

Introduction

1.1 Nuclear fusion: principle

The nuclear fusion is the reaction that happens in the core of the Sun and the other stars. The core is the central, hot (about 1.5 10^7K in the Sun) part in which is produced the radiated energy by the following nuclear reactions:

$^1H +^1 H \longrightarrow^2 D + e^+(positon) + 1MeV + \nu(0.42MeV neutrino)$
$^2D +^1 H \longrightarrow^3 He + \gamma(5.49MeV)$
$^3He +^3 He \longrightarrow^4 He + 2^1H + 12.86MeV$
which can be summed up in:
$4^1H \longrightarrow^4 He + 2e^+ + 2\nu + 26.7MeV$

The mass default generated by this reaction is converted into energy through γ rays. Every second, the sun converts 564 millions of tons of hydrogen into 560 millions of tons of helium. This 4 millions of tons mass default is what created the radiated energy by the sun.

It is interesting to notice that these reactions occur in a range of temperature that is above 10^7K. At these temperatures, gases are completely ionized and are in the form of plasmas. A plasma is an ionized gas respecting the quasi-neutrality property (the sum of the charges is equal to zero in a macroscopic volume) and behaving in such a way that the collective behavior is predominant over the individual behavior. The properties of the different plasmas are presented in figure 1.1.

The challenge for the controlled fusion is to understand the behavior and the properties of the plasma in order to be able to confine it and heat it for nuclear fusion.

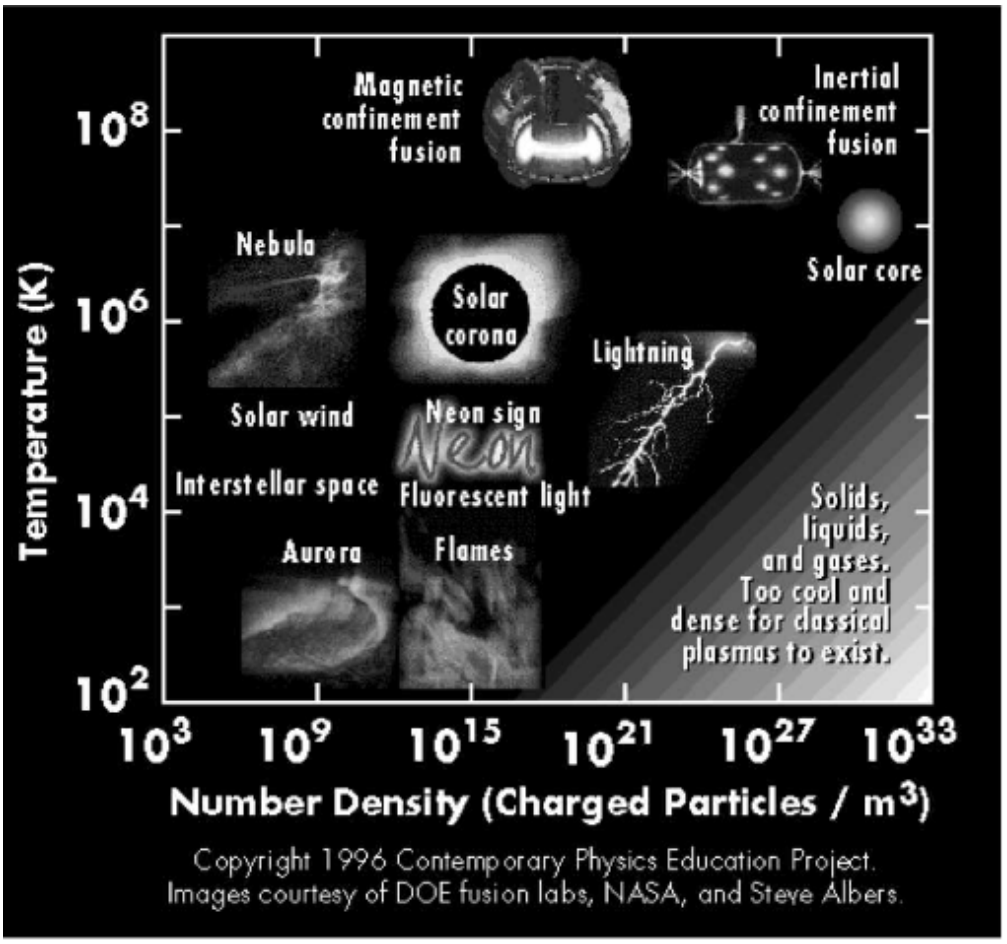


FIGURE 1.1: Typical plasma characteristics

1.2 Controlled fusion

Producing on earth conditions that are favorable to the nuclear fusion is a field that gathers a lot of scientific energy for mastering nuclear fusion could solve our energetic issues.

Therefore, several ways to achieve thermonuclear fusion on earth are being explored. The two main processes are the inertial confinement fusion and the fusion by magnetic confinement. We will focus on the fusion by magnetic confinement for the rest of the report.

The most relevant fusion reaction for the production of energy on earth are :

$$D + T \longrightarrow He(3.52MeV) + n(14.1MeV)$$
$$D + D \longrightarrow T(1.01MeV) + p(3.03MeV)$$
$$D + D \longrightarrow 3He(0.82MeV) + n(2.45MeV)$$

$$D + 3He \longrightarrow 4He(3.07MeV) + p(14.07MeV)$$

Nonetheless, these reactions only have a relevant cross-section at temperatures that are higher than 10 keV. These temperatures are higher than all the electron-core cohesion energy, which leads the gas to be fully ionized and therefore to exist only in the form of plasma.

If we have a closer look at the cross-section on figure 1.2, the maximum cross-section is obtained for the D-T reaction at a temperature of 100keV. For this reason, the deuterium and the tritium were privileged as a fuel.

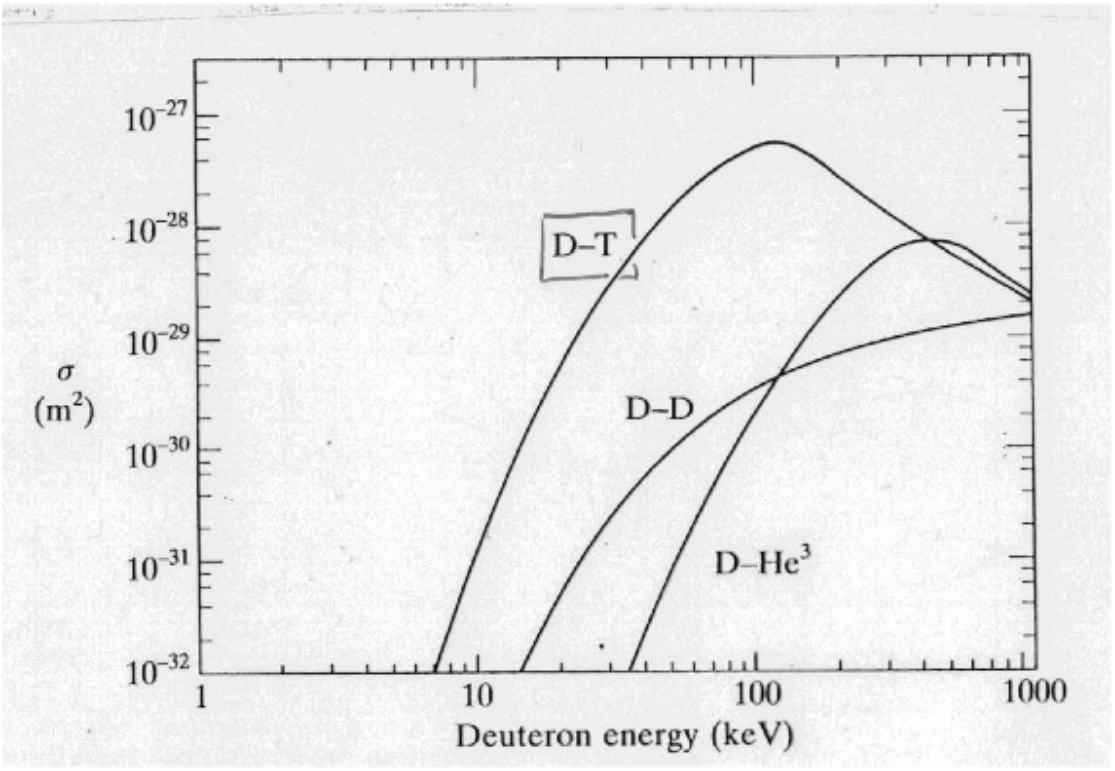


FIGURE 1.2: Cross-sections of the relevant fusion reaction (from "Tokamaks" by J.Wesson)

Now let's consider a plasma where the thermonuclear fusion reaction happens. It can be shown that in order to have a positive energy efficiency of the D-T reaction, the plasma characteristics must respect the Lawson criteria:

$$n_e T\tau > 5.1021 m^{-3}.keV.s \tag{1.1}$$

where n_e is the electron density, T the electron temperature and τ the confinement time. In order to increase the $n_e T\tau$ product, one can either increase n_e or T or τ. The temperature T is fixed by the reaction and is, for thermonuclear fusion, in the order

of magnitude of 10-20 keV. The inertial confinement fusion is playing on the electron density to improve the triple product, with $n_e \sim 10^{31} m^{-3}$ and $\tau \sim 10^{-10} s$.The thermonuclear fusion by magnetic confinement is on the opposite increasing the confinement time to up to 10s with an electron density of about $10^{20} m^{-3}$

1.3 Towards ITER

1.3.1 Presentation

The International Thermonuclear Experimental Reactor (ITER) is an international tokamak machine project that aims at proving the feasibility of a fusion based nuclear reactor. The success of such a project would lead to a cleaner and safer nuclear energy.

The principle characteristics of ITER are announced to be:

- Thermal power of the fusion: 500MW
- Electrical power used for the heating of the plasma: 50MW
- Energy balance: Q=10
- Major and minor radius: 6.20m and 2m
- Height of the plasma: 6.80m
- Volume of the plasma: $840 m^{-3}$
- Plasma current:15MA
- Toroidal magnetic field:5.3T
- Lifetime of the plasma: between 6min to 1h

1.3.2 Tokamak

A Tokamak is a magnetic confinement toroidal vessel used to control the plasma for nuclear fusion. The confinement's goal is to keep the plasma in a limited volume, far enough from any equipment to limit heat transfer, and thus any heat linked damage. An example of a tokamak is the TCV (Tokamak á configuration variable) based in CRPP, presented in figure **??**

The plasma being formed of charged particles, it is possible to confine their trajectories within a torus using magnetic fields. To achieve such a magnetic confinement, it is

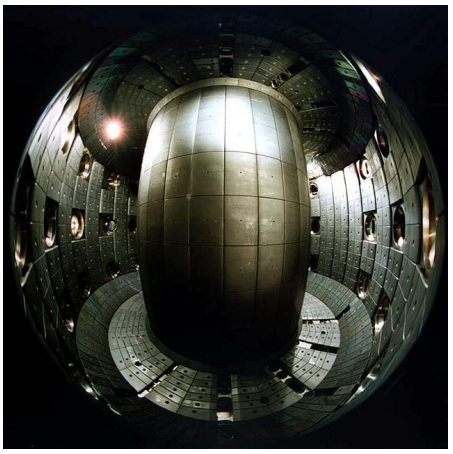


FIGURE 1.3: An indide view of TCV

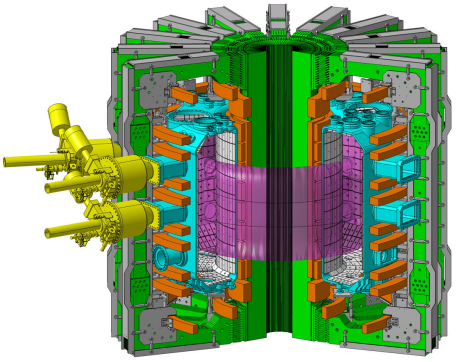


FIGURE 1.4: A schematic view of TCV:

FIGURE 1.5: An example of a tokamak: Tokamak á configuraiton variable (TCV) based in CRPP

necessary to to combine both a toroidal field, generated by toroidal coils, and a poloidal field, generated by an inducted current inside the plasma. Figure

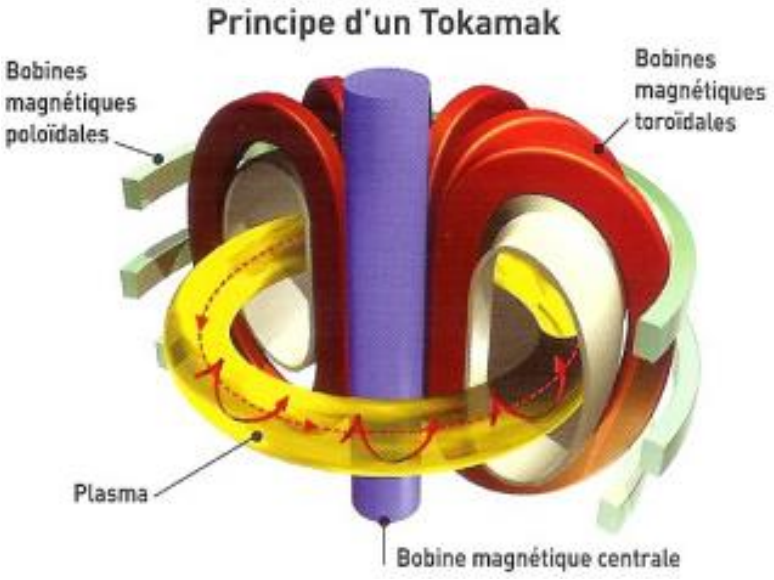


FIGURE 1.6: The principle of a tokamak:The red coil are responsible for the toroidal magnetic field as the green coils are responsible for the inducted current in the plasma. The trajectory of the center of mass is represented by the red arrows

1.3.3 The heating of the plasma

We have seen in section 1.2 that the required electron temperature for the thermonuclear fusion is $T_e = 10keV$. In [1], it is shown that the resistivity of the plasma is proportional

to $T_e^{-\frac{3}{2}}$ which means the hotter the plasma, the more difficult to heat. It is thus impossible to heat the plasma to the range of temperature where the nuclear fusion happens. Two external heating technics are used:

- The heating by neutral beam injection, heats the plasma by the collision between high energy neutrals and the plasma particles
- The heating by microwave injections in the plasma. The frequencies of the microwaves are chosen to correspond to the electron cyclotron or the ion cyclotron frequencies. We can see in figure **??** that the microwave launched from the external part of the vessel first encounters the turbulent Scrape Off Layer (SOL) of the plasma before interacting with the bulk plasma. It is thus important to know the effect of this turbulent layer on the propagation of the microwave to be able to control the heating by microwave. The understanding of the scattering of the microwaves by the SOL in a tokamak is the purpose of this study.

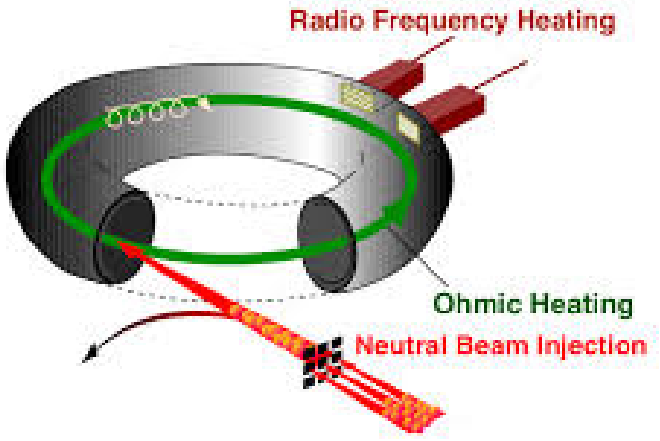


FIGURE 1.7: The three different technics for plasma heating

1.4 Fluid Model

1.4.1 Hypotheses

For the purpose of this study, we consider a cold, inhomogeneous, magnetized bounded plasma within the frame of the Fluid Model.

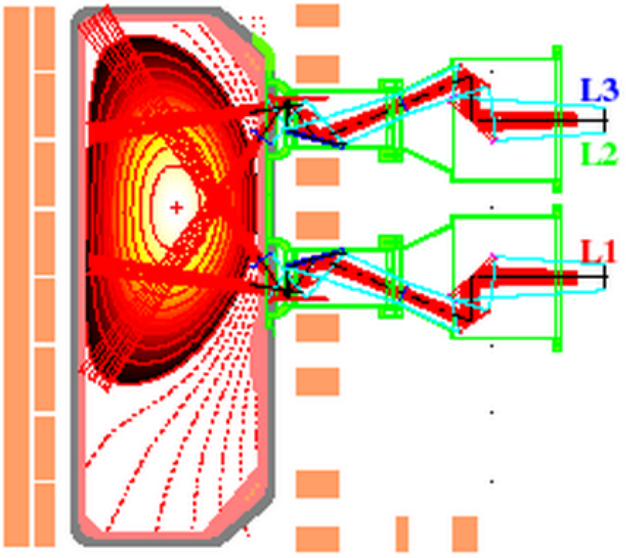


FIGURE 1.8: Microwave launchers in TCV

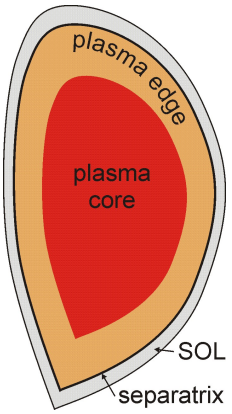


FIGURE 1.9: A poloidal section of the plasma

FIGURE 1.10: The radio-frequency waves used for the plasma heating go through the scrape off layer before reaching the bulk plasma

1.4.2 Fluid description of a plasma

Considering the plasma as a set of particles and solving the equation of motion for each one of them is a very complex problem that requires gargantuan amount of computer memory and calculus power. For many cases, the plasma observed in real experiments can be described by a simpler model. Within the Fluid Model, the identity of the particles is neglected and only the behavior of the fluid element is taken into account. The plasma can be described as a fluid composed of charged particles and is therefore capable of being influenced by electric and magnetic fields. A plasma composed of both ions and electrons (represented by the index k=i,e) can be described by the following equations:

Continuity equation:

$$\frac{\partial n_k}{\partial t} + \nabla.(n_k \mathbf{u}_k) = 0,\ k = i, e \tag{1.2}$$

Newton's equation:

$$n_k m_k [\frac{\partial \mathbf{u}_k}{\partial t} + (\mathbf{u}_k.\nabla)\mathbf{u}_k] = -\nabla p_k + n_k q_k (\mathbf{E} + \mathbf{u}_k \times \mathbf{B}) + \sum_{k \neq l} \mathbf{R}_{kl},\ k = i, e \tag{1.3}$$

The term $\mathbf{R}_{kl}$ represents the coulombian collisions between the different species with

$$\mathbf{R}_{kl} = n_k m_k \nu_{kl} (\mathbf{u}_l - \mathbf{u}_k) = -\mathbf{R}_{lk} \tag{1.4}$$

Maxwell's equations:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \tag{1.5}$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}, \tag{1.6}$$

One may have more precision on the establishment of the fluid model for the plasma by having a look at [2].
Within the frame of our study, the plasma is considered to be cold, therefore the pression term $-\nabla p_k$ is negligible. Moreover, the typical time scale of the waves supported by the plasma being small compared to the collision times, we consider that $\mathbf{R}_{kl} = 0$. Also, at the time scale of the RF waves, we consider the ions to be motionless and only study the dynamics of the electrons. We can then write 1.7 as:

Newton's equation:

$$n_e m_e [\frac{\partial \mathbf{u}_e}{\partial t} + (\mathbf{u}_e.\nabla)\mathbf{u}_e] = n_e q_e (\mathbf{E} + \mathbf{u}_e \times \mathbf{B}) \tag{1.7}$$

1.4.3 Solving the wave equation

The problem of the scattering of an RF (Radio Frequency) wave by turbulent structures in the plasma means solving solving the inhomogeneous wave equation. Let's consider the configuration as shown in figure 1.11

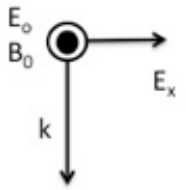


FIGURE 1.11: Configuration of the propagation

From the Maxwell's equations, we can derive that:

$$\nabla(\nabla \mathbf{E}) - \nabla^2 \mathbf{E} = -\mu_0 \frac{\partial \mathbf{j}}{\partial t} - \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t} + Boundary\ conditions \tag{1.8}$$

which is, in the frequency domain:

$$\nabla(\nabla\widetilde{\mathbf{E}}) - \nabla^2\widetilde{\mathbf{E}} = -\mu_0\omega\mathbf{j} - \frac{\omega^2}{c^2}\widetilde{\mathbf{E}} + \textit{Boundary conditions} \tag{1.9}$$

knowing that $\mathbf{j} = \sigma\mathbf{E}$, we have:

$$\nabla(\nabla\widetilde{\mathbf{E}}) - \nabla^2\widetilde{\mathbf{E}} = \frac{\omega^2}{c^2}\epsilon_r\widetilde{\mathbf{E}} + \textit{Boundary conditions} \tag{1.10}$$

with

$$\epsilon_r = \frac{i\sigma}{\epsilon_0\omega} \tag{1.11}$$

ϵ_r can be expressed from the linearized dynamic equations from the electron population within the fluid model and assuming a cold plasma and no collision with neither the ions nor the neutrals. Further details can be found in **??**

$$\epsilon_r = \begin{bmatrix} \epsilon_1 & -i\epsilon_2 & 0 \\ i\epsilon_2 & \epsilon_1 & 0 \\ 0 & 0 & \epsilon_3 \end{bmatrix} \tag{1.12}$$

where

$$\epsilon_1 = 1 + \frac{\omega_{pe}^2}{\Omega_e^2 - \omega^2} \tag{1.13}$$

$$\epsilon_2 = -\frac{\Omega_e}{\omega}\frac{\omega_{pe}^2}{\Omega_e^2 - \omega^2} \tag{1.14}$$

$$\epsilon_3 = 1 - \frac{\omega_{pe}^2}{\omega^2} \tag{1.15}$$

ω_{pe} representing the electron plasma frequency and Ω_e the electron cyclotron frequency. ω_{pe} and Ω_e can be expressed as follow

$$\omega_{pe} = \sqrt{\frac{e^2 n_e}{\epsilon_0 m_e}} \tag{1.16}$$

$$\Omega_e = \frac{eB_0}{m_e} \tag{1.17}$$

where e is the electric charge of the electron, n_e the electron density, ϵ_0 the dielectric permittivity of the vacuum, m_e the electron mass and B_0 the external magnetic field.

1.4.3.1 Dispersion relations

if we introduce R and L such as ϵ_1=R+L and ϵ_2=R-L, the dispersion relation for the *ordinary (O) mode* is reduced to

$$n^2 = \epsilon_3 \tag{1.18}$$

where $n^2 = \frac{c^2k^2}{\omega^2}$ is the index of refraction

The dispersion relation for the *extraordinary (X) mode* is

$$n^2 = \frac{RL}{\epsilon_1} \tag{1.19}$$

1.4.4 Numerical model

The partial differential equation (PDE) 1.9 is solved in COMSOL using the finite element method. For further detailed information on the method and its application in electrodynamics, one may refers to [3].

Grid As a first application, the problem is solved on a 2D triangular grid which size can be user-controlled[1]. For the following, we have set the maximum element size to the 1/8 of the incident wave wavelength. A convergence study on a test case is presented in section 2.1.3.

1.4.4.1 Boundary conditions

In the following, **n** refers to vector normal to the interface plasma/boundary. According to the hypotheses used in the simulation, the boundary conditions can be defined as follow.

[1] COMSOL can also be used to solve the problem in 3D

Radiation Conditions Radiation conditions are used when we need to simulate an unbounded domain. In that case, the electric and magnetic field are required to satisfy the *Sommerfeld radiation condition.*

$$\lim_{r\to\infty} r(\nabla \times \begin{pmatrix}\mathbf{E}\\ \mathbf{H}\end{pmatrix} + jk_0\mathbf{r} \times \begin{pmatrix}\mathbf{E}\\ \mathbf{H}\end{pmatrix}) = 0 \tag{1.20}$$

where $r = \sqrt{x^2 + y^2 + z^2}$ which are valid if $r >> |a|$ with $|a|$ being the characteristic size of the scatterer.

These radiation conditions express the idea that the radiated energy must scatter to infinity and that no energy can come back from infinity to the domain of interest.These radiation conditions are explained in details in [3].

For cases in which the inequality $r >> |a|$ cannot be satisfied, more sophisticated radiation boundary condition are needed for taking into account the higher-order multi polar component of the radiating field. In COMSOL, this condition is satisfied using a Scattering Boundary Condition.

Perfectly Matched Layer (PML) Another way to simulate an infinite space in a bounded region is to use Perfectly Matched Layers (PML) as shown in figure 1.12. Perfectly matched layers were first developed by Berenger in 1994. A PML is a layer of absorbing material that is placed at the edge of the grid that is independent of the boundary conditions. When a wave enters a PML,it decays exponentially, if it is reflected at the boundary, the returning wave is exponentially tiny. Berenger also showed that the wave does not reflect at the interface between the region of interest and the PML. However, when it comes to using PMLs in numerical simulations, some features can become real limitations. First, as soon as the problem is discretized for numerical issues, reflections may appears for one is only solving the approximate wave equation, the analytical perfection of the PML is no longer valid. Then, the absorption of the PML is angle-dependent. The attenuation rate is proportional to $|k|\cos\theta$ where $|k|$ is the norm of the wavevector and θ the angle between the wave vector and the PML. It appears clearly that when the radiation approaches the glancing incidence, the attenuation rate of the PML goes to zero, which means that all the corresponding energy is reflected. One should be very careful with those limitations while using the PMLs.

A comparison between the efficiency of the Scattering boundary conditions and the PML will be done in chapter 2.

Perfect Electric Conductor The perfect electric conductor boundary condition is

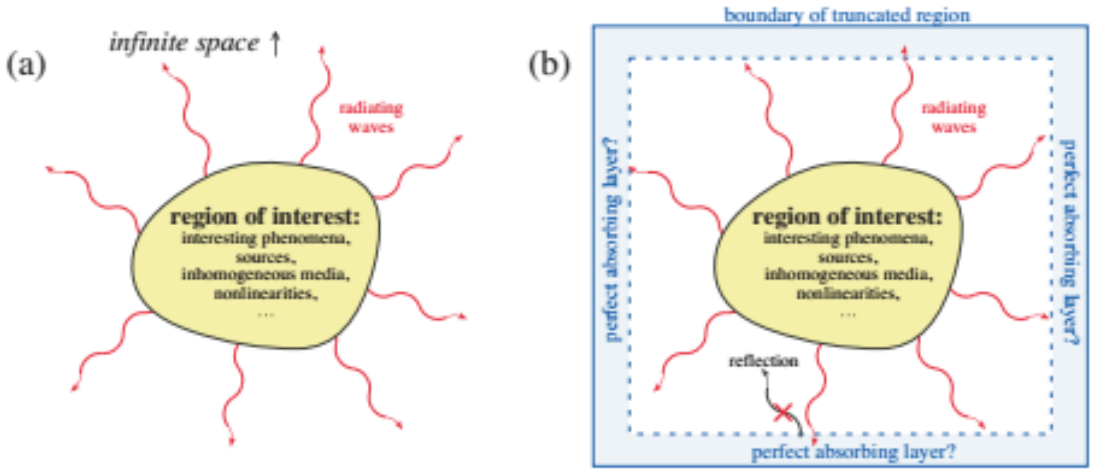


FIGURE 1.12: from [4] : (a) Schematic of a typical wave-equation problem, in which there is some finite region of interest where sources, inhomogeneous media, nonlinearities, etceteras are being investigated, from which some radiative waves escape to infinity. (b) The same problem, where space has been truncated to some computational region. An absorbing layer is placed adjacent to the edges of the computational regiona perfect absorbing layer would absorb outgoing waves without reflections from the edge of the absorber.

$$\mathbf{n} \times \mathbf{E} = 0 \tag{1.21}$$

It is a special case of the electric field boundary condition that sets the tangential component of the electric field to zero. It is used for the modeling of a loss less metallic surface when the electric conductivity $\sigma_c \to \infty$.

Impedance Electric Conductor In order to model a real electric conductor, a more appropriate boundary condition is the Impedance Electric Conductor boundary condition as described in [3].

in that case, it can be shown that the electric and magnetic field are related by:

$$Z_0 \sqrt{\frac{\mu_r}{\epsilon_c}} \mathbf{n} \times \mathbf{H} + \mathbf{E} - (\mathbf{n}.\mathbf{E})\mathbf{n} = 0 \tag{1.22}$$

where $Z_0 = \sqrt{\frac{\mu_0}{\epsilon_0}}$ μ_0 being the dielectric permeability of the vacuum,μ_r the dielectric permeability of the conductor, ϵ_0 the dielectric permittiviy of the vacuum and ϵ_{rc} the relative complex dielectric permittivity of the conductor, including the loss term.ϵ_{rc} can be written as $\epsilon_{rc} = 1 + \frac{\sigma_c}{i\omega\epsilon_0}$ where σ_c is the electrical conductivity of the conductor, i the imaginary number and ω the pulsation of the incident wave. For further information on these impedance electric conduction boundary conditions, one may refer to [5]

Chapter 2

COMSOL Validation and predictions for the TORPEX experiment

2.1 COMSOL validation

COMSOL is a commercial tool used, in our case, for solving the wave equation using the finite elements method. Before using such a tool, it is essential to test the validity of the solutions computed by COMSOL and to be able to spot the limitations of the simulation.

In this section, we will first compare the COMSOL simulation for the scattering of an RF wave by a metallic cylinder with the analytical simulation for different boundary conditions. Then, we will analyse the reaction of the plasma simulated by COMSOL to the injection of an RF plane wave and compare it to the theory.

2.1.1 Scattering of an RF wave by a metallic cylinder : Comparison between theory and COMSOL

The goal of this study is to understand the influence of the boundary conditions on the results calculated by COMSOL. The frame of the study is represented on figure 2.1 . A normal incidence plane wave on the TM^z polarization is scattered by a perfectly conducting circular cylinder. We compare the results obtained for the zcomponent of the electric field obtained analytically and using COMSOL for different boundary conditions.

The electric field was solved analytically in [6]. It is shown that :

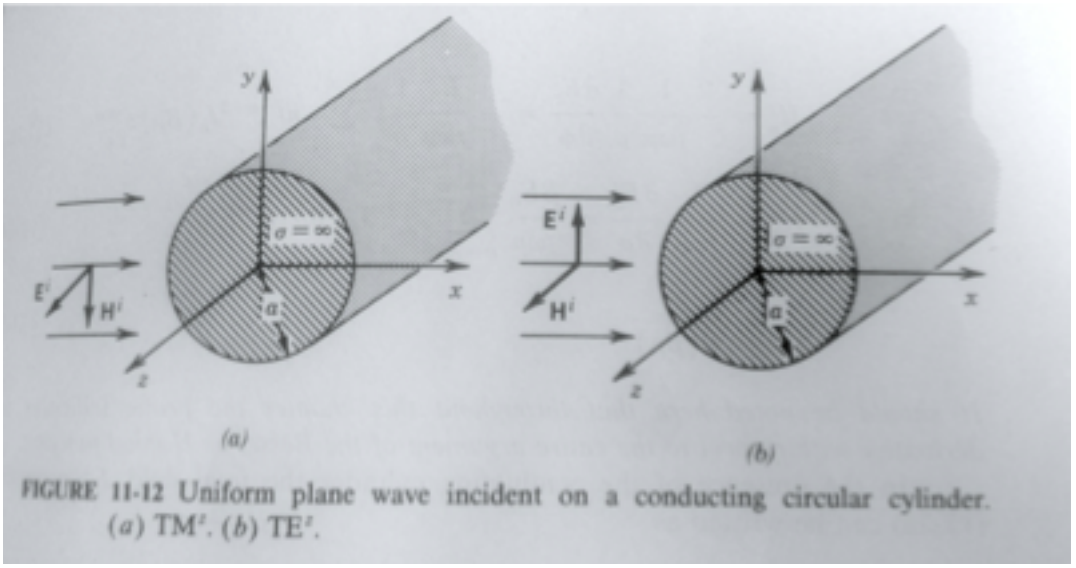


FIGURE 11-12 Uniform plane wave incident on a conducting circular cylinder. (*a*) TM^z. (*b*) TE^z.

FIGURE 2.1: scattering of an RF wave by a conducting cylinder, setup of the study (from []Balanis89

$$E_z = E_0 \sum_{n=-\infty}^{+\infty} j^{-n}[J_n(\beta\rho) - \frac{J_n(\beta a)}{H_n^{(2)}(\beta a)} H_n^{(2)}(\beta\rho)]e^{jn\phi} \tag{2.1}$$

here E_z is the total field, E_o the amplitude of the incident field, j the imaginary number, J_n the first type Bessel function, $H_n^{(2)}$ the second type Hankel function, β the wave vector, a the radius of the cylinder and ρ and ϕ the cylindrical coordinates. We set a=1mm and a frequency of the incident wave f=20Ghz and f=50Ghz such that we have two different $\frac{\lambda}{a}$ ratios, λ being the wavelength of the incident plane wave.

Studying the convergence of the equation 2.1 The calculus of the electric field was implemented in Matlab using the equation 2.1. The convergence of the serie in 2.1 being very slow, we first analyse it in figure 2.1

The results presented in figure 2.2 were computed for f=20Ghz. We can notice that after n=20, the convergence is such that it is hard to perceive the influence of the terms corresponding to larger ns. For this reason, we choose n=20 for the rest of the study when f=20Ghz. Nonetheless, for f=50Ghz, the ratio $\frac{\lambda}{a}$ being smaller, the convergence of the serie can be considered as good only for n greater than 50. In that case, we set n=50.

Comparing analytical results with matlab simulations The area of calculus is a square which size is 3cm. The boundary conditions on the sides parallel to the

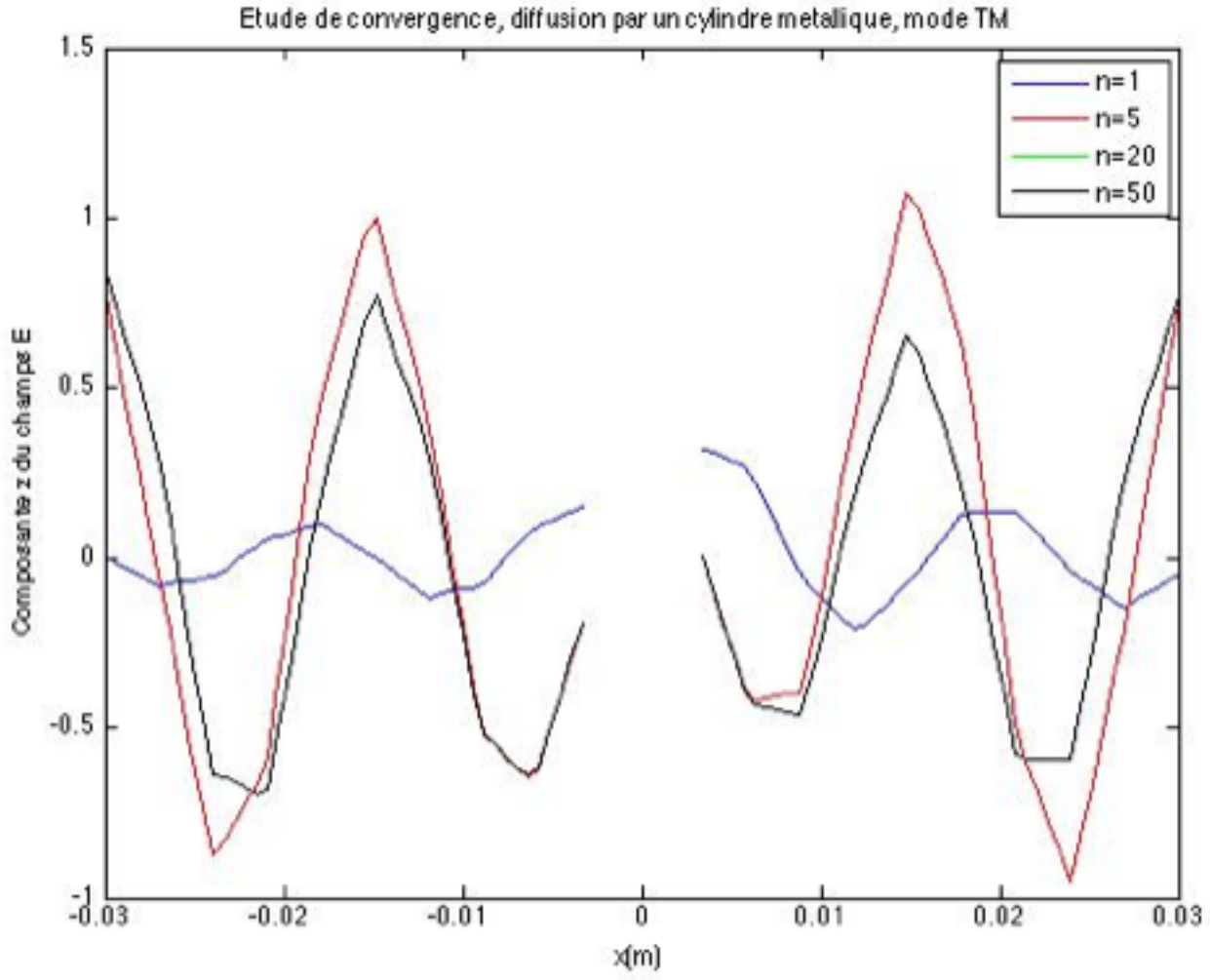


FIGURE 2.2: Convergence of the analytical solution of the equation **??**

propagation are PMLs and scattering boundary conditions on the side perpendicular to the propagation. Figure 2.6 and 2.10 give the result of the comparison between theoretical results and COMSOL results for f=20Ghz and f=50Ghz. We can see that for a bigger $\frac{\lambda}{a}$ ratio, the correspondence of the results is better, this is due to the fact that for such a ratio, the convergence of 2.1 is better.

Influence of the boundary conditions Even for a simple plane wave propagating in the vaccuum, we can notice that the COMSOL side effects are perceptible on the propagation of the wave as shown in figure 2.13. In order to study the influence of the boundary conditions, we change the profile of the incident wave to be Gaussian, with a profile such that it can, with a good approximation be considered as a plane wave around the cylinder.

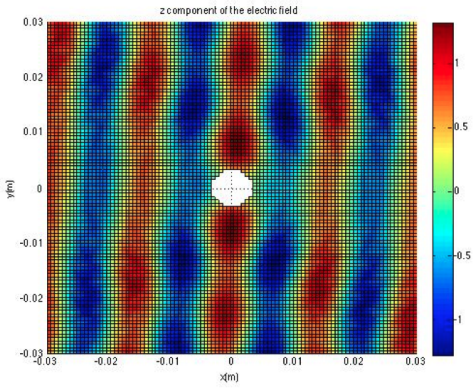


FIGURE 2.3: Analytical solution

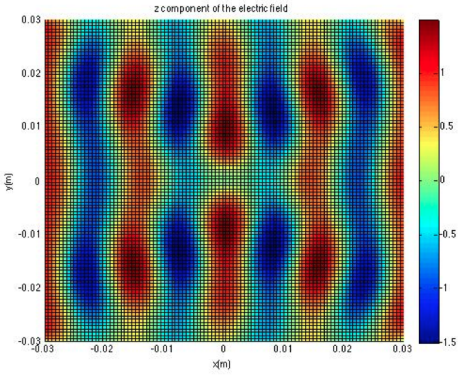


FIGURE 2.4: COMSOL simulation

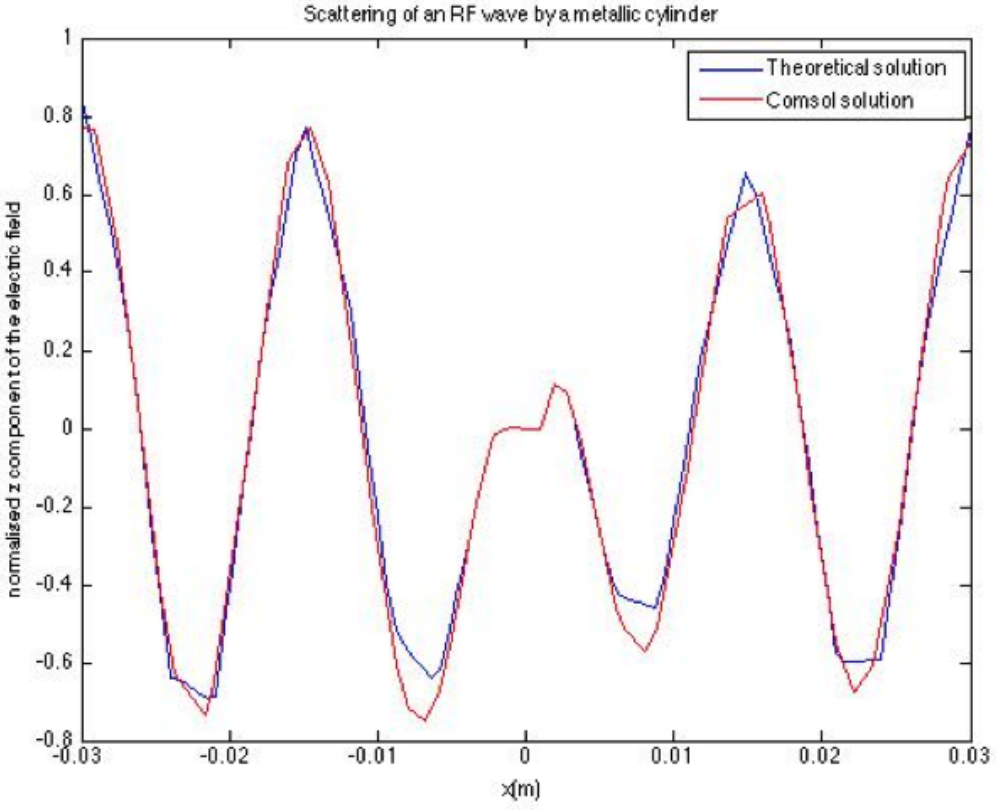


FIGURE 2.5: Horizontal profil for E_z along y=0m

FIGURE 2.6: Comparing COMSOL and theoretical results for E_z normalized to the incident electric field, f=20GHz, a=1mm

We can see that the propagation of the Gaussian beam avoids having an electric field with a wave vector parallel to the PMLs and therefore reduces the side effects in a very considerable way. We then change the geometry to be cylindrical in order to avoid having wave vectors parallel to the boundaries of the calculus area. We compute the scattering of the RF wave by the cylinder for f=50Ghz with different boundary conditions : PMLs and scattering boundary conditions. The results are presented in figure 2.14. We can

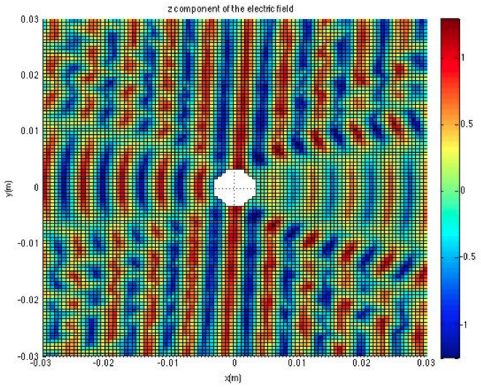


FIGURE 2.7: Analytical solution

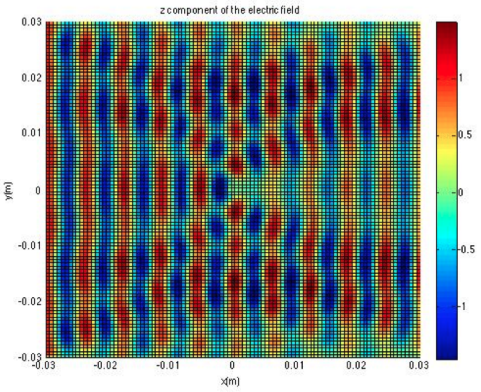


FIGURE 2.8: COMSOL simulation

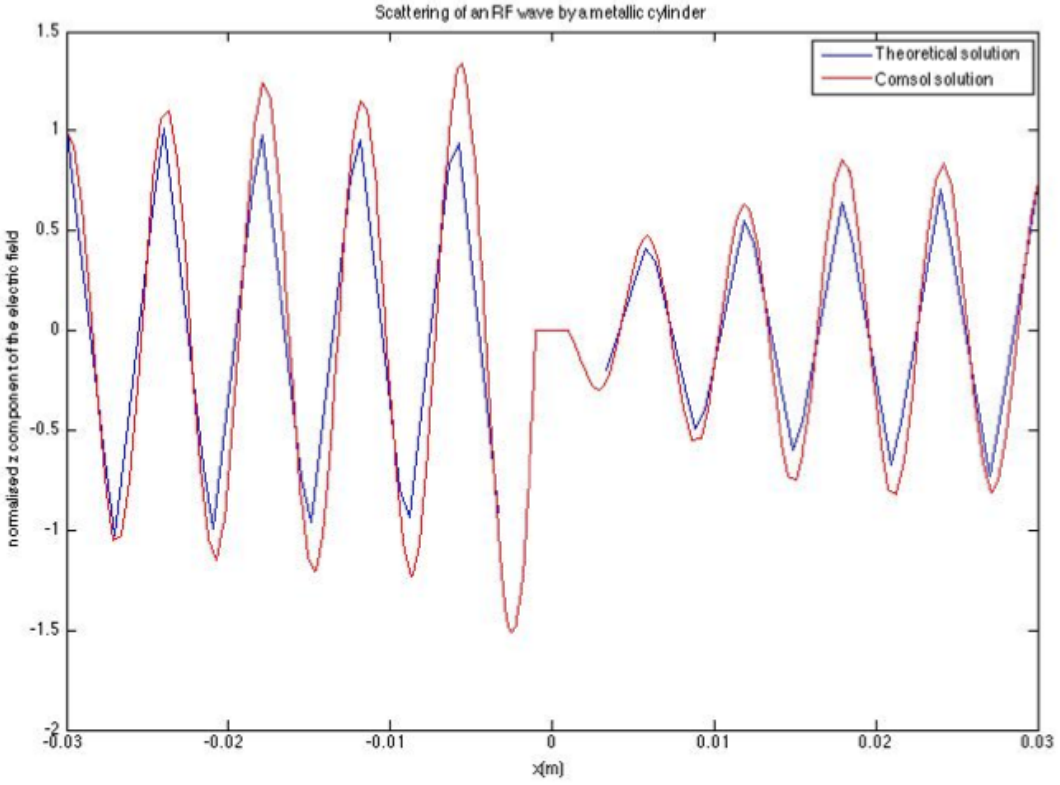


FIGURE 2.9: Horizontal profil for E_z along y=0m

FIGURE 2.10: Comparing COMSOL and theoretical results for E_z normalised to the incident electric field f=50GHz, a=1mm

see that the most satisfying solution is the one corresponding the the propagation of the Gaussian beam and also that their is no considerable difference, in the case of a cylindrical geometry between a PML or scattering boundary conditions.

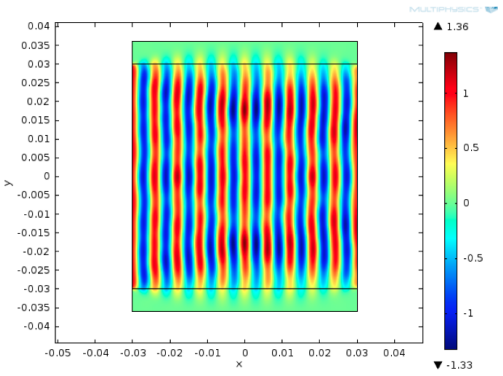


FIGURE 2.11: Plane wave

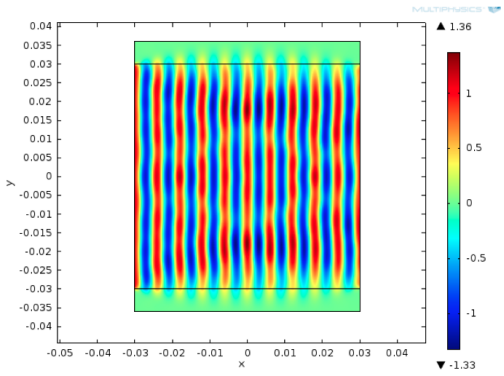


FIGURE 2.12: Gaussian beam

FIGURE 2.13: Effects of the PML boundary conditions on the propagation of a plane wave and a Gaussian beam in the vacuum on COMSOL

2.1.2 Implementing the plasma dielectric tensor in COMSOL

COMSOL enables the user to define new materials by their physical properties. We thus implement the dielectric tensor 1.12 to define the plasma in COMSOL. In order to validate plasma response to an electromagnetic plane wave, we run the following simple simulations: A plane wave is propagating at the interface vacuum/plasma 2.15.

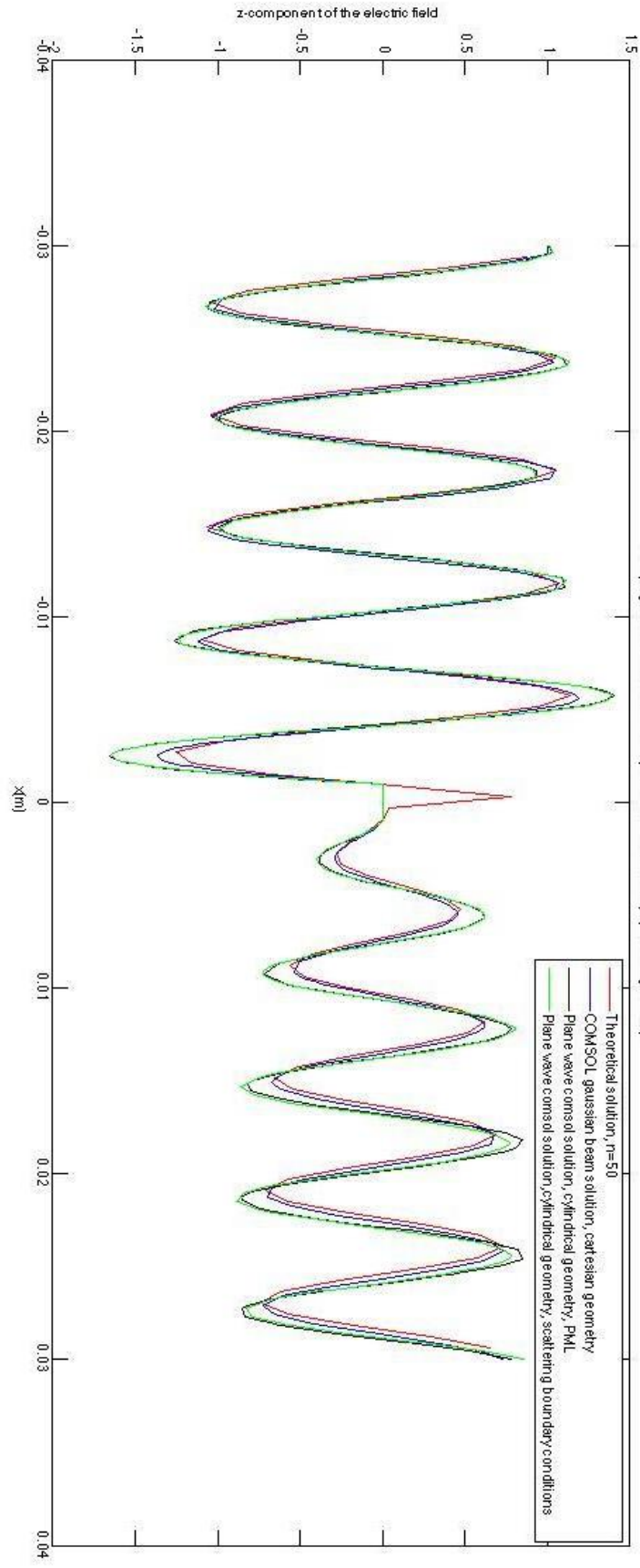
We first report the calculated values of ϵ_r in COMSOL and compare them with the theoretical prediction. The two values correspond perfectly as shown in figure 2.16

We therefore explore the behavior of the system at particular densities. To do so, we make a section of a certain component of the electric field along the propagation. The section is made along the dashed line represented in figure 2.15. For instance, let's consider the X-mode and analyze the electric field. By sweeping the electron density from 10^{19} to 2.10^{20} $m^{-}3$ we can clearly see the upper-hybrid resonance at $n_e = 7.10^{19} m^{-3}$ since the wave polarization at the UH resonance satisfies the ES nature ($\mathbf{E}//\mathbf{k}$) which implies a strong E_y component which is shown in Fig2.17.

Now if we focus on an electron density such that it is beyond the cut-off and beyond the UH resonance, for instance $n_e = 2.10^{20} m^{-3}$, we can see that the wave is vanishing into the plasma as reported in figure 2.18.

2.1.3 Computing time

This chapter comes as an additional piece of information. Regarding the computational effort while running the validation tests, we notice a strong influence of the mesh grid on the calculation time. On the local computer which is an iMac intel core i5, 2.5Ghz, 4Go RAM, we ran the previous case at $n_e = 3.10^{19} m^{-3}$ for different sizes of the mesh


z-component of the electric field
x(m)
studying the effect of the boundary conditions on Comsol, cylindrical geometry
Theoretical solution, n=50
COMSOL gaussian beam solution, cartesian geometry
Plane wave comsol solution, cylindrical geometry, PML
Plane wave comsol solution,cylindrical geometry, scattering boundary conditions

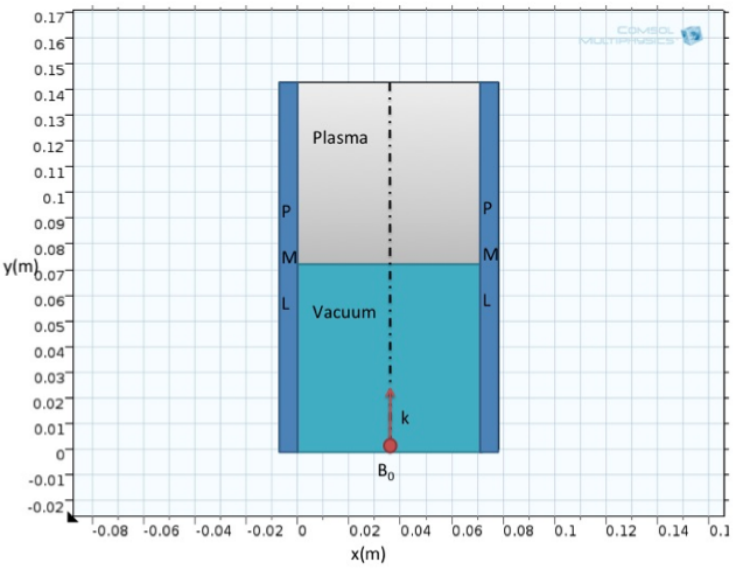


FIGURE 2.15: The geometry used for the validation tests: a plane wave propagates at the interface vacuum/plasma

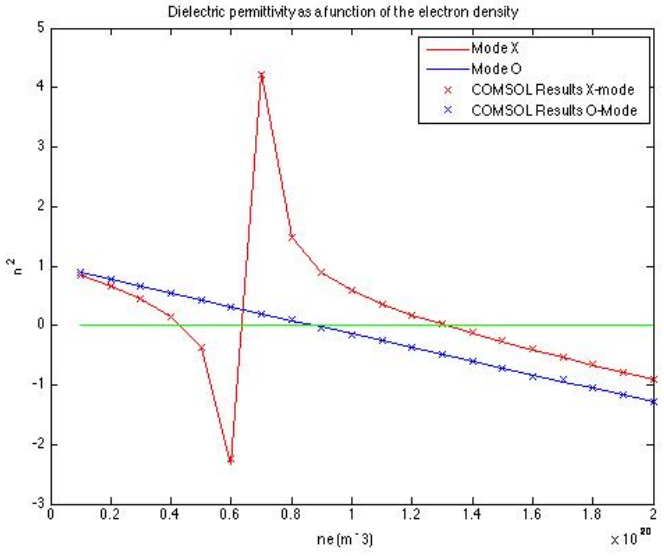


FIGURE 2.16: the comparison between the calculated value of ϵ_r through COMSOL and the theoretical value correspond perfectly.

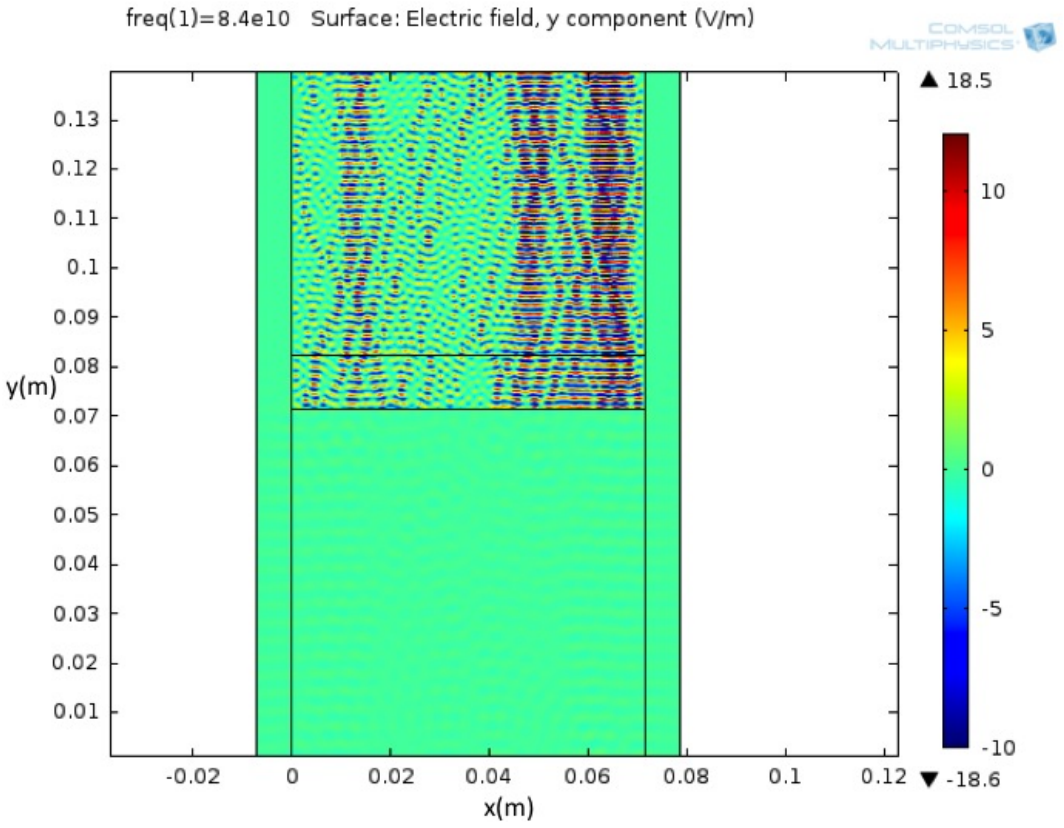


FIGURE 2.17: y component of the electric field at the upper-hybrid resonance

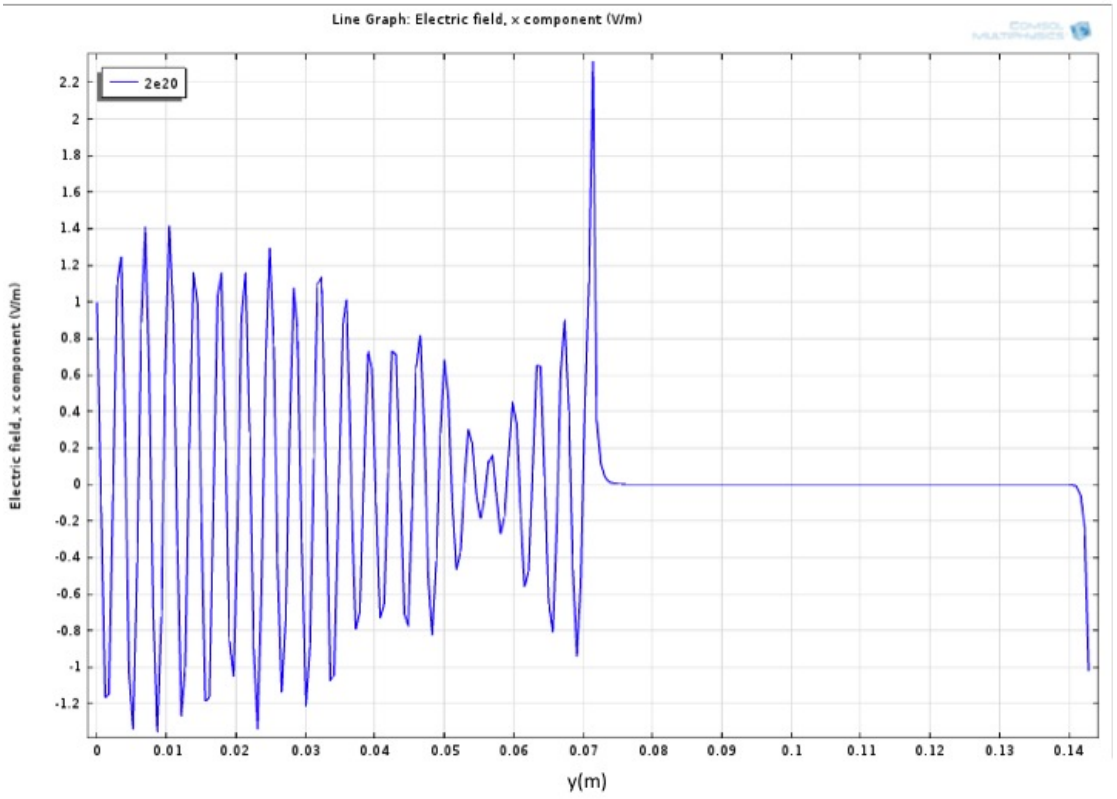


FIGURE 2.18: A vanishing wave into the plasma. x component of the electric field

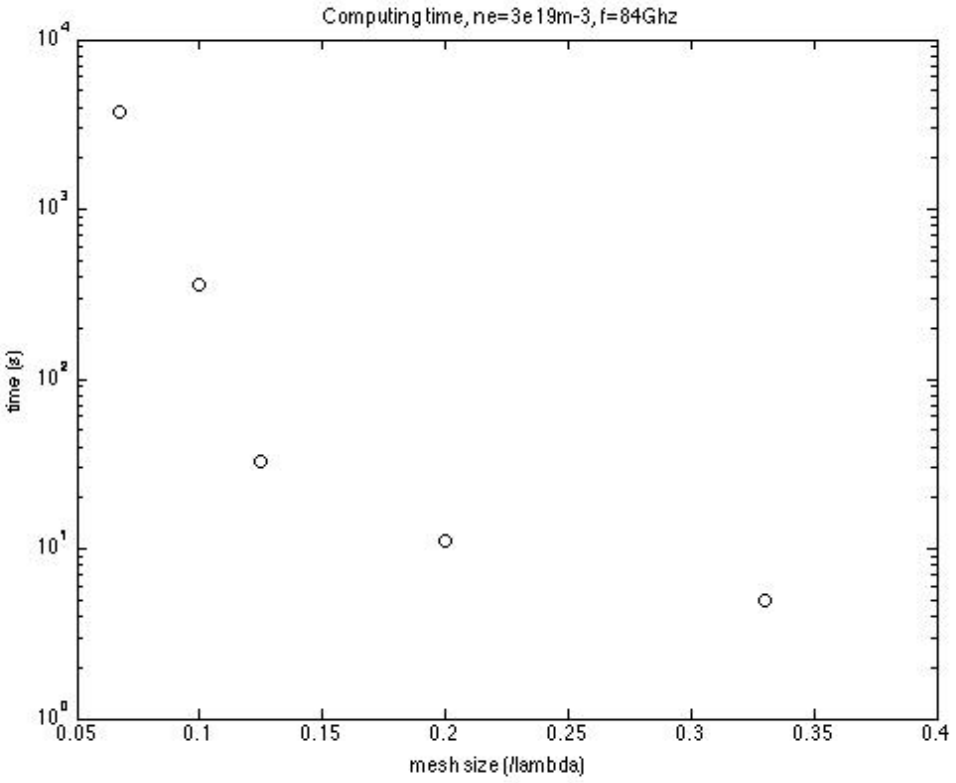


FIGURE 2.19: Calculation time as a function of the mesh elements' size

elements. These sizes are represented as a fraction of the incident wavelength and the results is reported in figure 2.19. The scale is semi-logarithmic on the calculation times. We can see that the calculation time varies exponentially with the mesh elements size. For a good comparison of the effect of the mesh size on the quality of the simulation results, a zoom on the vanishing area from the of the x component of the electric field for $n_e = 5.10^{19}$, figure 2.21, is presented in figure 2.22. It is good to point out that the electron density has no relevant effect on the computational time.The memory resources used for the simulation are also dependent on the mesh size as shown in figure 2.20. It is therefore very important to find a compromise for each simulation between the expected resolution of the result,the calculation time and the available computational res sources.

2.2 COMSOL predictions for the experimental campaign

Using a TORPEX geometry and a TORPEX typical plasma configuration, we run some very simple 2D simulations to visualize the propagation of an RF wave in the X-mode at f=10GHz in the TORPEX vessel, without plasma, with plasma in the HFS and plasma in the middle (the plasma profiles are represented in figure 2.23. We analyze the detection in the X-mode, nothing being detected in the O-mode. We find out that a very complex

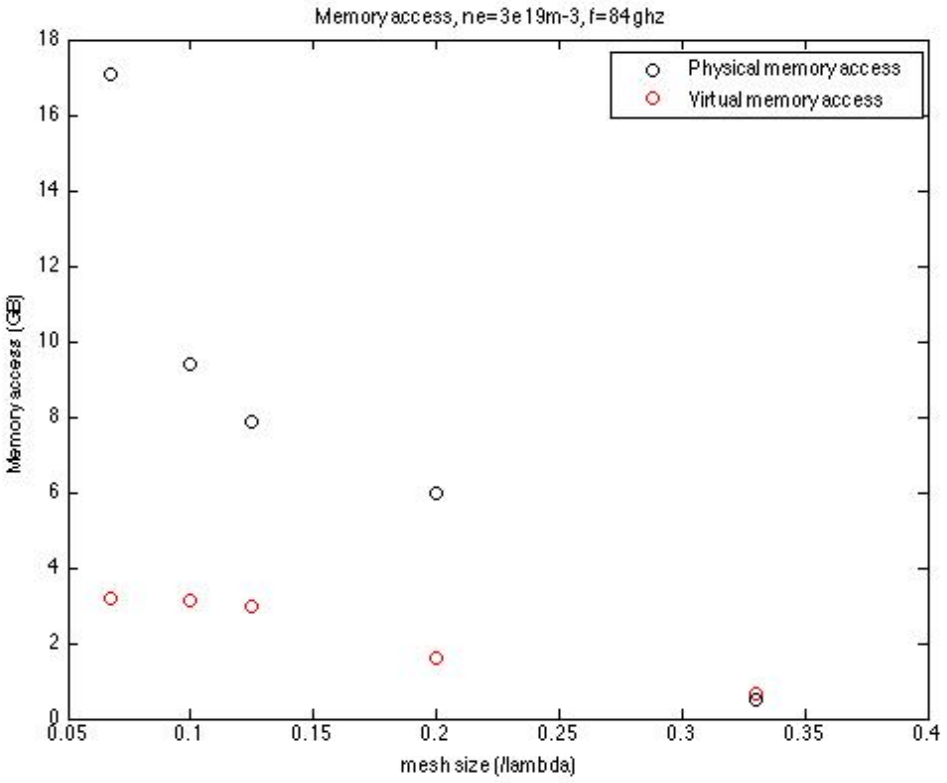


FIGURE 2.20: Memory access as a function of the mesh elements' size

mode distribution is created in the TORPEX vessel that is influenced by the presence or not of the plasma (figure2.24) but that does not seem to be influenced by the position of the plasma (figure 2.25)

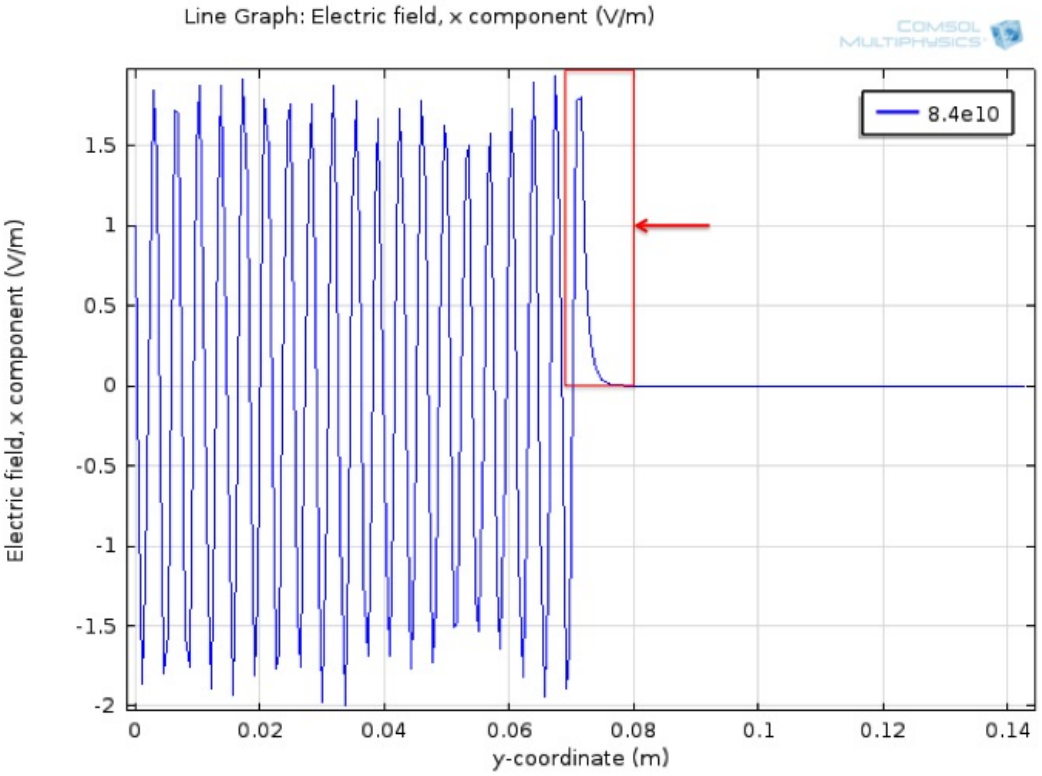


FIGURE 2.21: A vanishing wave into the plasma. x component of the electric field $n_e = 5.10^{-19}$. The zoom in figure 2.22 corresponds to the area within the red rectangle.

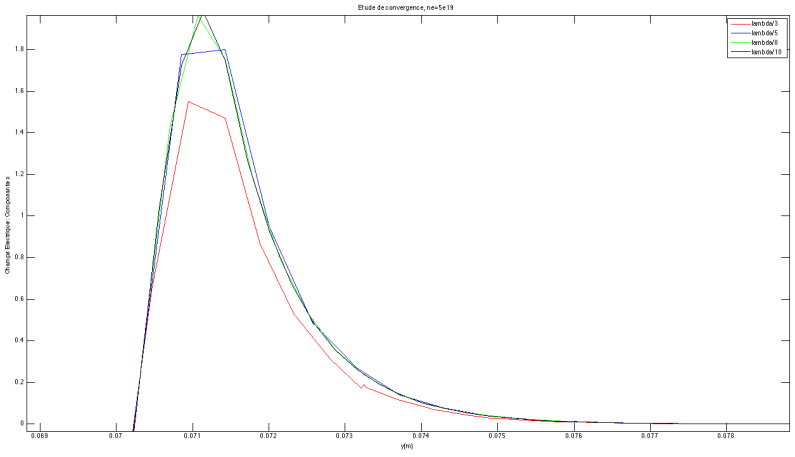


FIGURE 2.22: Comparison of the effect of the size of the mesh elements. Zoom on figure 2.21

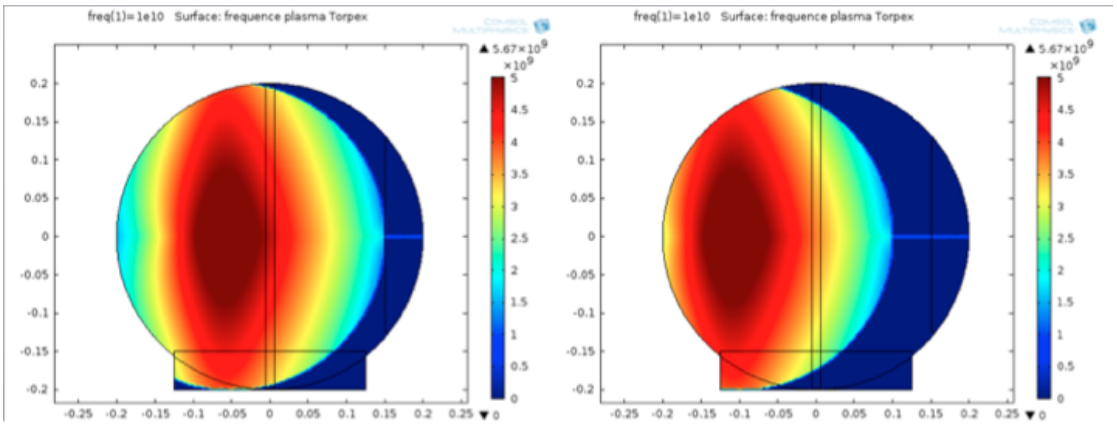


FIGURE 2.23: Two TORPEX profiles: a plasma in the HFS (left) and a plasma in the middle (right). The quantity represented is the electron density in m^{-3}

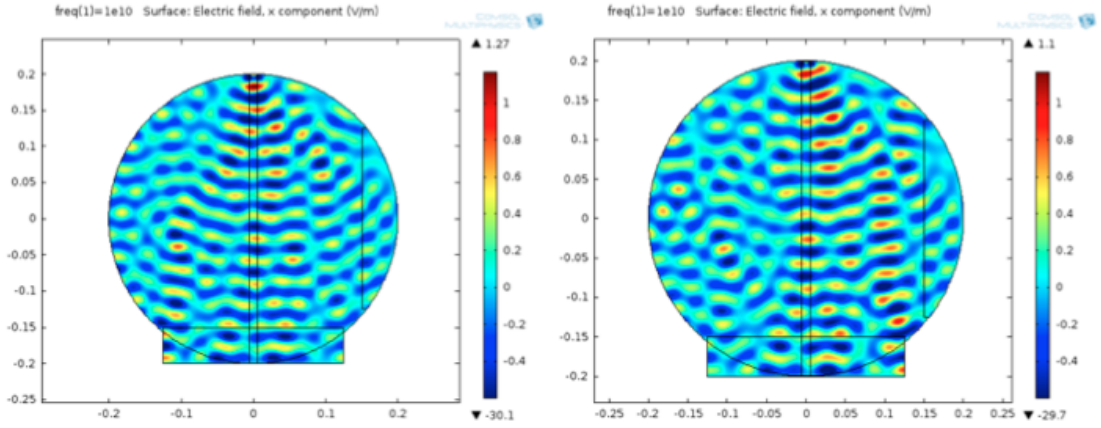


FIGURE 2.24: Comparison of the E_x with and without plasma: the presence of the plasma disturbs the modes in the vessel

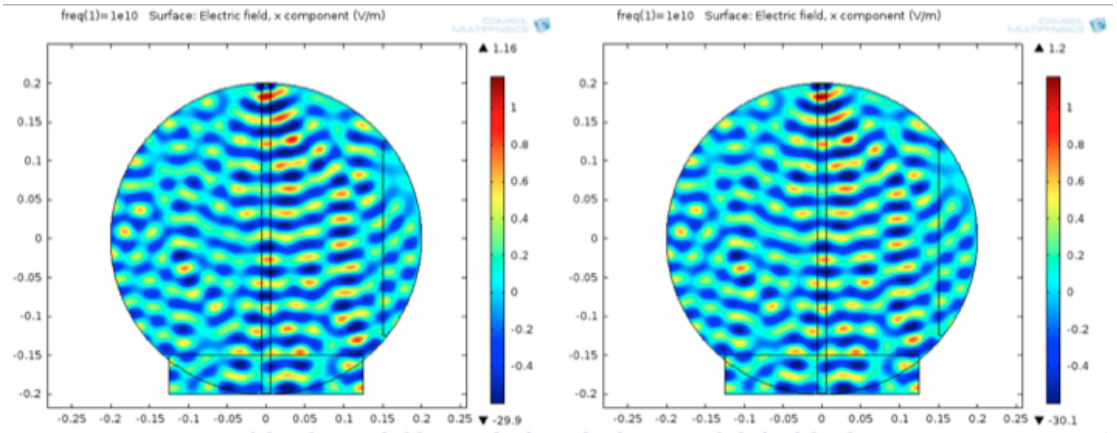


FIGURE 2.25: Comparison of the E_x for two plasma profiles (plasma in the HFS and plasma in the middle: the position of the plasma does not seem to disturb the modes of the vessel

Chapter 3

Experimental Setup

In this chapter, the TORPEX device is presented as well as the RF diagnostic together with some possible improvements. We will also insist on the Conditional Averaged Sampling (CAS) data analysis technique.

3.1 The TORPEX device

The TORPEX device is presented in figure 3.1 The TORPEX vacuum vessel has major and minor radius of 1m and 0.2m. The horizontal coil allow for a vertical field up to 5mT while the toroidal coils generate a toroidal field up to 0.1T. The plasma is produced by a microwave source of 2.45GHz, which corresponds to the electron cyclotron range of frequencies. The microwaves are injected perpendicular to the plasma in the low field side (LFS). The existing free electrons are accelerated at the electron cyclotron frequency and the plasma if formed by the impact ionization. The wave is injected in the O-mode to avoid the cut-off of the X-mode. For the purpose of this study, a H plasma is made with typical electron density in the range of $10^{16} - 10^{17} m^{-3}$ and an electron temperature in the range of 5-10 eV.

3.2 The RF diagnostic

3.2.1 Diode-based RF diagnostic

The RF diagnostic is presented in figure ???. It consists of an injection and a detection system. The injection system is made of an RF tunable source, between 6 and 20Ghz with an associated line width of about 100MHz, and a rectangular waveguide in the X

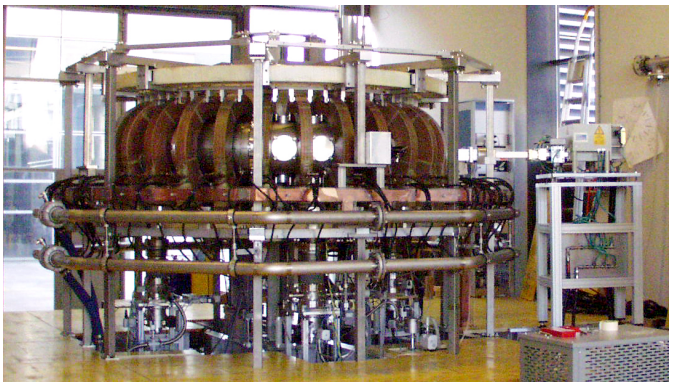


FIGURE 3.1: a view of the TORPEX expertiment

band (8-12GHz). We limit the injected wave frequency to 12.5 GHz to only excite the guide mode TE_{10}. The waveguide is mounted on movable system that enables both choosing the injection mode, which will be set for the rest of the study to the X-mode, and the position of injection, which will be set to r=11cm in the TORPEX Coordinates. The detection system is made of a rectangular waveguide in the X band connected to a Schotcky diode (HP 3333D, with a bandwidth of 33GHz). Both the injected power and the detected power can be visualized on an acquisition channel that has a frequency acquisition of 250kHz. We will therefore focus our study to the low frequency phenomena ($\leq 125kHz$). It is interesting to note that due to the waveguide size (1.11x2.22cm), it is not relevant to perform radial scans with a step smaller than 1cm. The whole setup is presented in figure 3.5.

This diagnostic enables to access the power collected by the waveguide which is proportional to the square of the electric field. Using a heterodyne detection system would enable to access the electric field component along the TE_{10} waveguide mode as well as the phase shift.

3.2.2 A possible improvement: A heterodyne-based detection system

An RF source G1 delivers a signal at 10GHz that is sent from the rectangular waveguide w1 to w2 and collected by the Mixer 2. C1 represents the TORPEX vessel. a second RF source, G2, acts like the local oscillator and has a frequency of 9.5Ghz. Mixer1s output IF1 is here as a phase reference for the measurement of the phase of IF2. The signals collected on the Lecroy oscilloscope is numerically low-pass filtered with a cutoff at 1Ghz, eliminating the higher frequency, enabling us to focus on the signal component of interest, the one at 500Mhz. More detailed explanation on the working of the mixers can be found in Microwave Engineering by David M.Pozar, Addison-Wesley, 1990. Amplitude

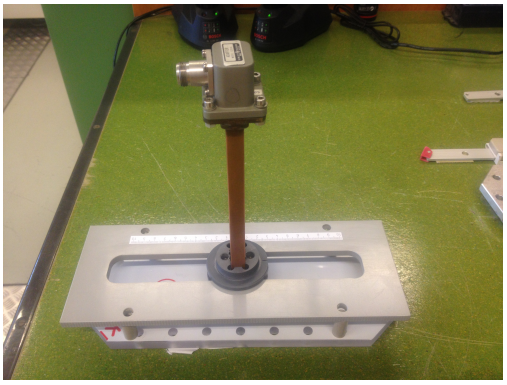


FIGURE 3.2: The RF mechanical support of the waveguide

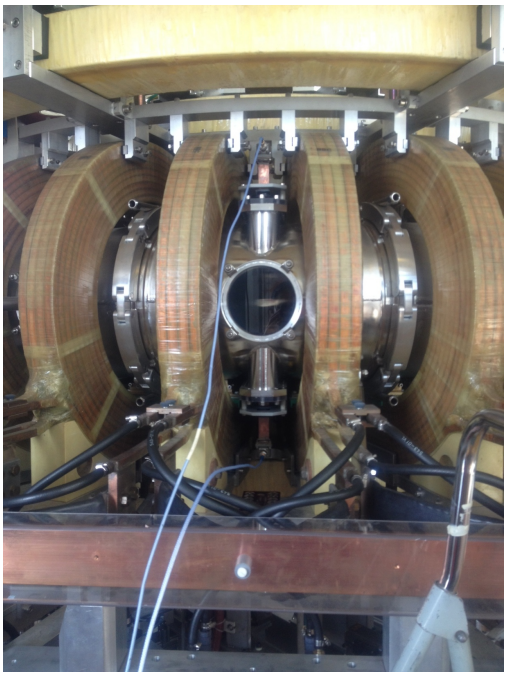


FIGURE 3.3: The RF diagnostic mounted on TORPEX

FIGURE 3.4: A visualization of the RF diagnostic

and phase measurements can be directly made on the oscilloscope, on a signal averaged over several sweeps to minimize the error.

3.3 Conditional Averaged Sample (CAS) technique

The CAS is a method used to extract coherent structures in turbulent plasmas. Let's assume a fluctuating equilibrium that consists of an incoherently fluctuating part together with intermittently occurring coherent structures, fast growing modes. The CAS enables to find and averaged temporal evolution of the structure. One probe, detects the occurrence of a coherent structures and thus define the trigger condition. For every trigged event, a sequence of a given length is scanned for a simultaneously taken temporal signal. The result is then averaged over all the trigged events.

An illustration is given in figure 3.7

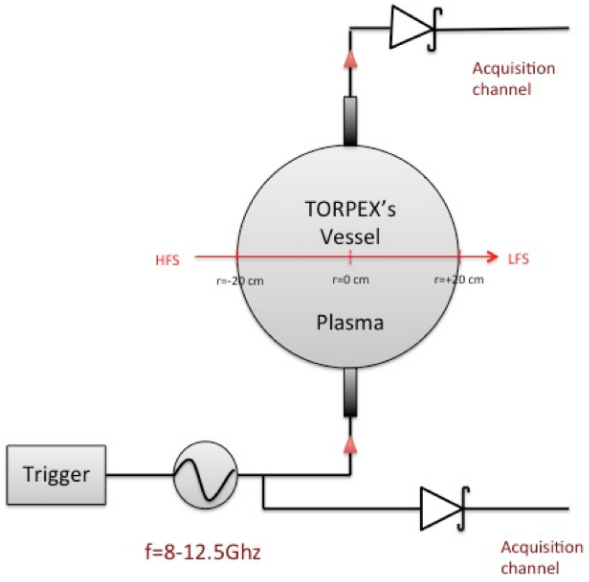


FIGURE 3.5: RF diagnostics: An RF wave at a frequency in the range 8-12.5 Ghz in the X-mode is injected from the bottom at r=+11cm and is detected at a position r that is variable in either the X or the O-mode.

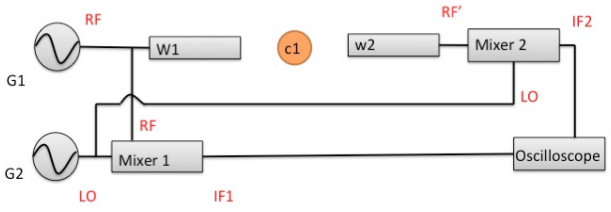


FIGURE 3.6: RF diagnostics: using a heterodyne system enables to access both the electric field and the phase shift

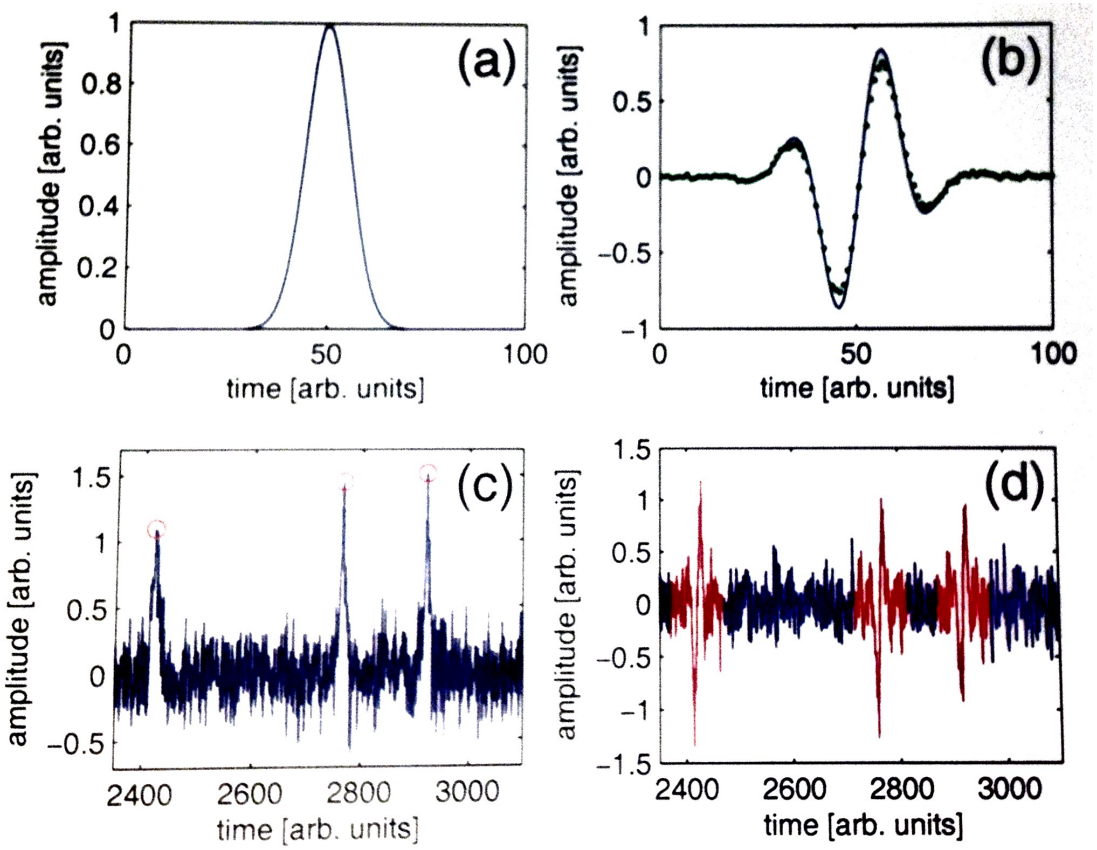


FIGURE 3.7: from [7]: figure (a) represents a coherent event on the reference probe and is detected to be used as a trigger in (c). the corresponding coherent event, in the signal of interest, is obtained in figure (b) averaging the sequences in red in (d) corresponding to the events.

Chapter 4

Experimental study of the RF scattering by the turbulent structures in the TORPEX plasma

4.1 Description of the plasma profiles

We want to study the interaction of the RF waves with the TORPEX Plasma. The preliminary COMSOL simulations presented in section 2.2 predicted a complex mode structure within the TORPEX vacuum vessel which may change according to the plasma profile in the vessel. We thus create three different plasma profiles presented in figure 4.1 which will respectively be referred to as High Field Side (HFS), middle and Low Field Side (LFS) plasma.

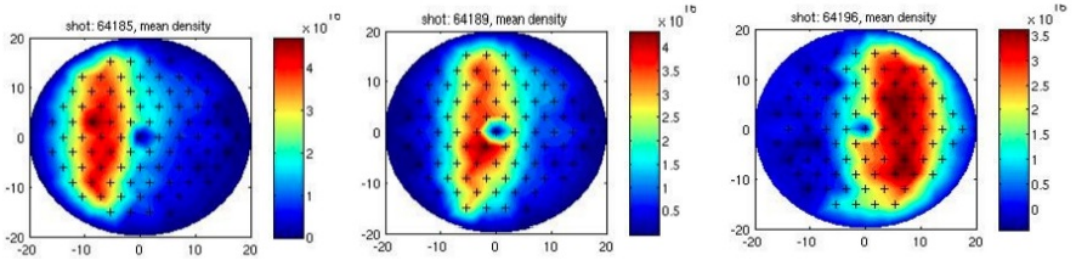


FIGURE 4.1: Mean electron density (in m^{-3} for the three plasma profiles (the hole in the middle correspond to a missing HEXTIP probe). From left to right, the plasma profiles will respectively be referred to as: High Field Side (HFS), middle and Low Field Side (LFS) plasma.

4.2 Mode structure in the TORPEX vessel

After having characterized the three plasma profiles, a radial scan using the RF diagnostic presented in 3.2 was performed with a detection in both the X and O-mode for f=8,12.5GHz.

The measurement of the power linked to the electric field in 4.2 suggest the presence of a global mode structure in the TORPEX vessel. As we can see in figure 4.9 and 4.2, the presence of the plasma modifies the global mode structure in the vessel in both the X and the O-mode. The plasma profile however does not seem to have such a significant effect on the mode structure.

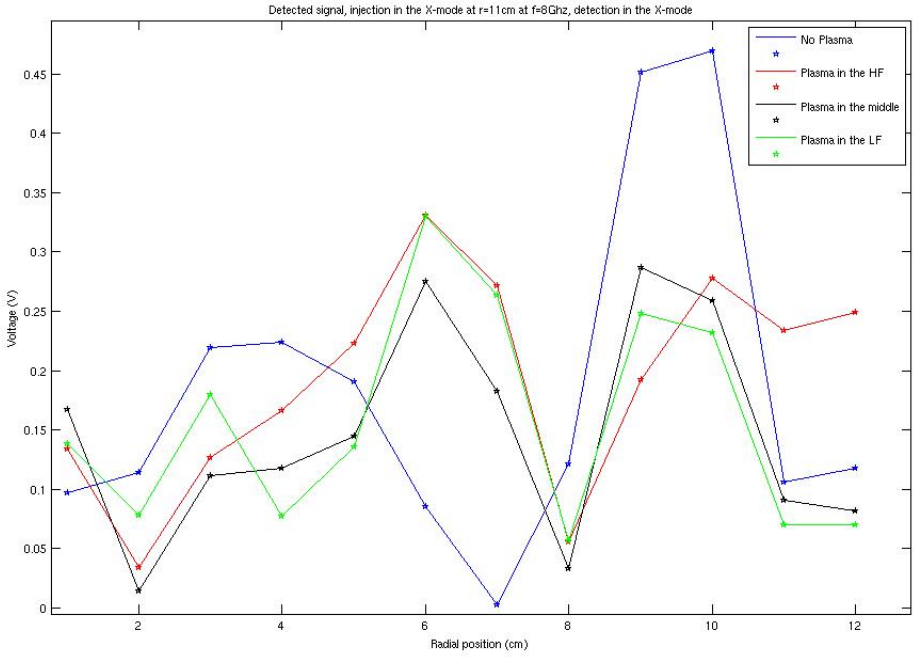


FIGURE 4.2: Comparison of the radial scan of the RF power detected in the X-mode, for different configurations: without plasma, plasma in the HFS, plasma in the middle and plasma in the LFS

4.3 Interaction between the plasma modes and the RF wave

The second noticed effect of the plasma on the RF signal after the effect on the mean value is the change in the standard deviation of the signal. Introducing a plasma in the

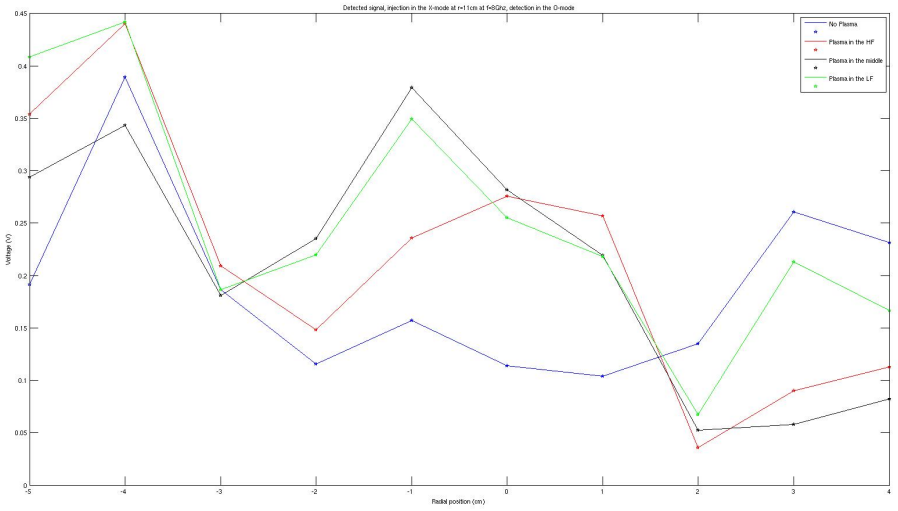


FIGURE 4.3: Comparison of the radial scan of the RF power detected in the O-mode, for different configurations: without plasma, plasma in the HFS, plasma in the middle and plasma in the LFS

TORPEX vessel immediately leads to an increase in the standard deviation if the RF signal as shown in figure 4.6

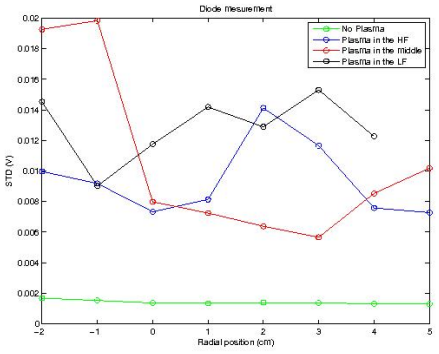


FIGURE 4.4: Detection in the O-mode

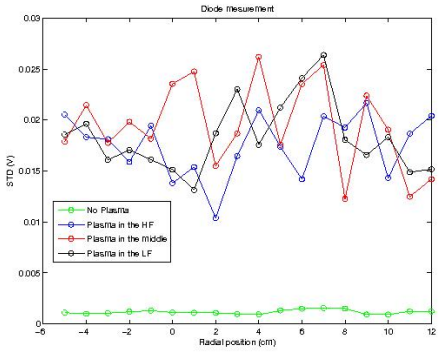


FIGURE 4.5: Detection in the X-mode

FIGURE 4.6: Standard deviation of the RF signal for a detection in both the X and the O mode at f=8GHz

Figure 4.6 also shows that the nature of the plasma has a strong influence on the fluctuations of the signal. To go further on the analysis of the interaction between the plasma modes and the RF waves we focus on the radial position r=-2cm where the standard deviation of the RF signals corresponding to the situations with plasma where the standard deviation is both the largest and the more different.

The power spectral density of the signal detected in the O-mode at both 8 and 12.5 GHz shows an evidence of the interaction between the RF wave and the plasma modes. The

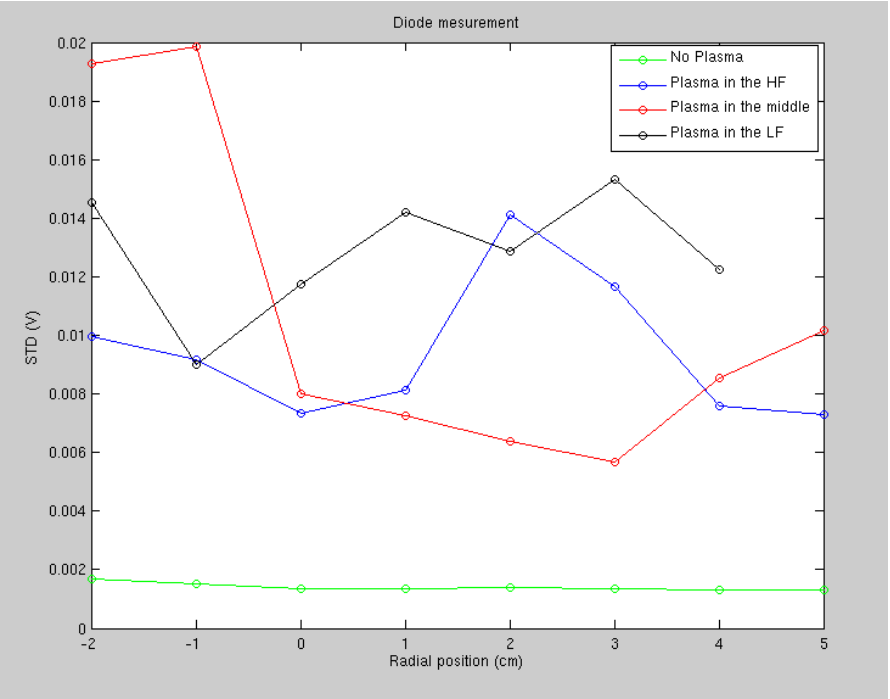


FIGURE 4.7: The introduction of the plasma in the cavity introduces a large standard deviation of the RF signal

peak frequency depends on the plasma position and can vary of a factor two (difference between the plasma in the middle and the other configurations) Other interesting features, yet unexplained, can be seen at around 20kHz for the LFS plasma and between 60-100kHz for the middle plasma. The peaks at 27GHz and its harmonics are caused by the magnetron power supply. The transparency of the plasma to an RF wave increasing with the frequency, we can see that the peaks corresponding to the plasma modes are much lower at 12.5GHz in comparison to the peaks at 27GHz.

To confirm the hypothesis of the peaks at around 2-6kHz corresponding to the interaction between the RF signal and the plasma modes, we compute on the one hand the correlation between the HEXTIP probes and the RF signal and on the other hand the cross spectral power density. Not only are the RF signal and the HEXTIP probes well correlated, as we can see in figure 4.10, but they also share most of their power between 2-6 kHz, depending on the plasma position. It is also interesting to notice that the correlation between the RF signal and the HXTIP probes reveals a 10.5ms delay between the two signals.

If we repeat the same exercise for the detection in the X-mode, we can see that the interaction between the plasma modes and the RF wave in the X-mode is almost non-existent (4.11 and 4.12) The peak at 2-6kHz disappears and the correlation between the two RF and HEXTIP signals is weak. The wave being injected in the X-mode, there

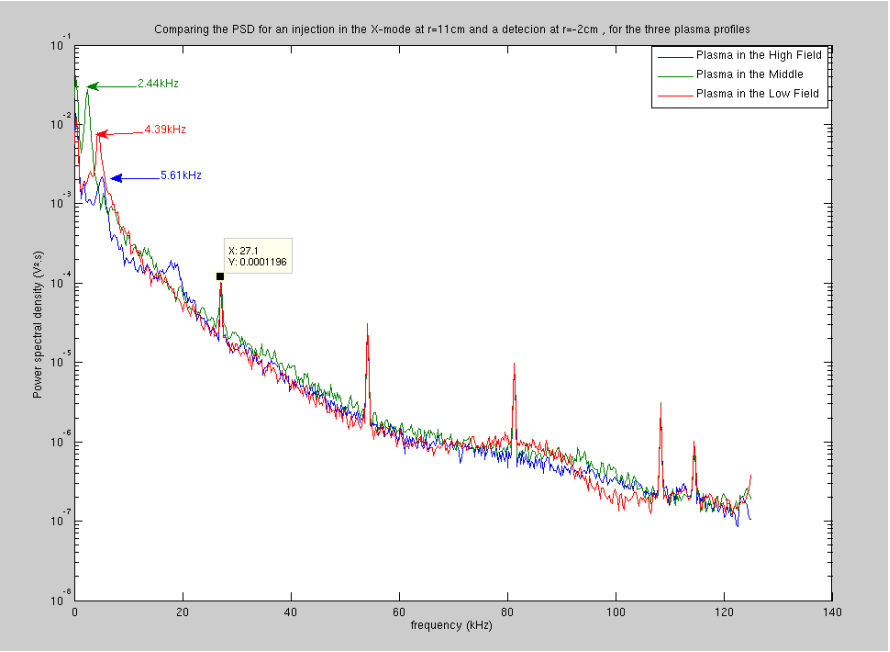


FIGURE 4.8: The spectral power density of the RF signal shows evidence of the interaction between the RF wave and the plasma modes around 2-6 kHz

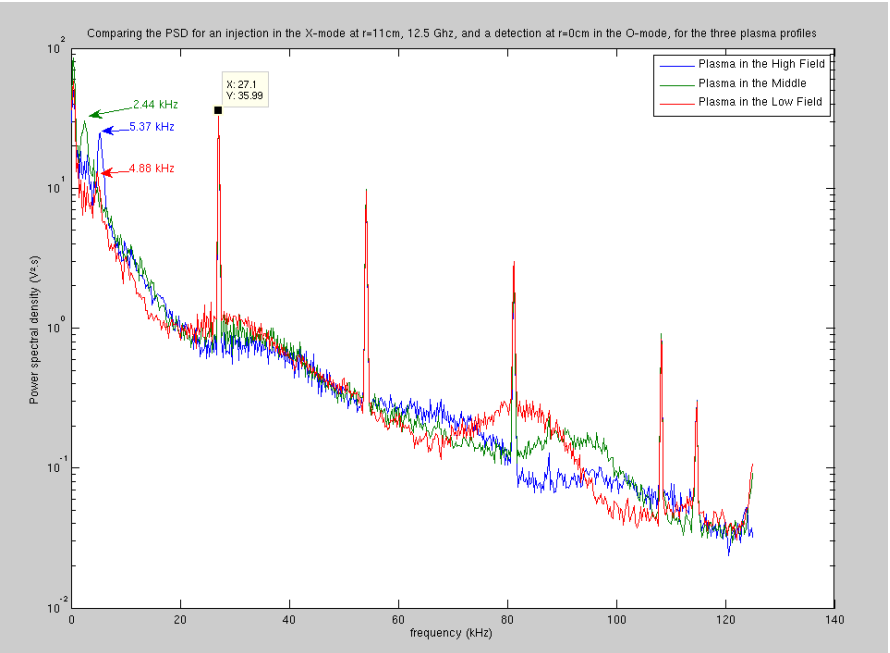


FIGURE 4.9: The spectral power density of the RF signal shows evidence of the interaction between the RF wave at 12.5 and the plasma modes around 2-6 kHz, the interaction is weaker than at 8GHz due to the increasing transparency of the plasma with the frequency

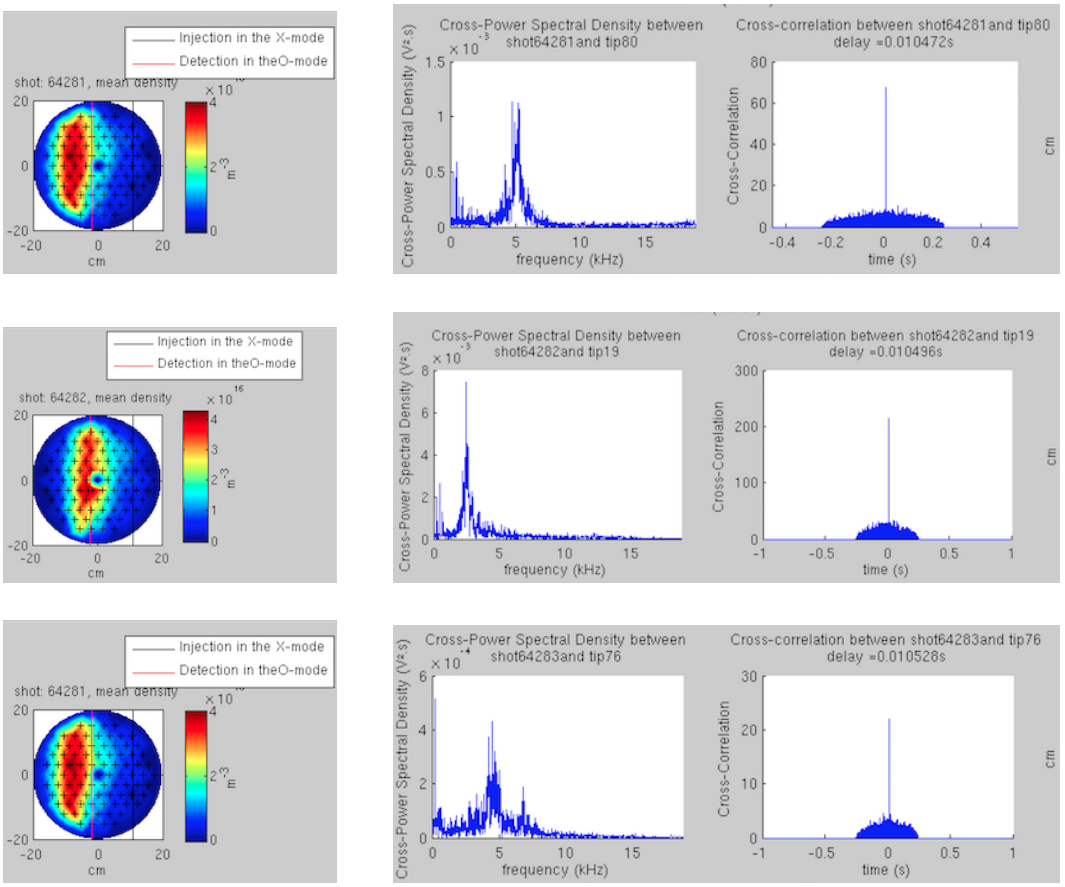


FIGURE 4.10: the correlation and the cross power spectral density were computed between the RF signal and the best correlated probe for the three different plasma profiles, a detection in the O-mode and a frequency of 8GHz.

are several hypothesis that could explain such a result. The first one is that the toroidal field being aligned with the helical field lines, it is not perfectly perpendicular to the TORPEX section, which means that while injecting in the X-mode, a slight part of the RF power is injected in the O-mode. This small component nonetheless has the greatest interaction with the plasma modes. Another hypothesis, yet more unlikely to happen in a TORPEX of plasma, is that the plasma is responsible for an X to O mode conversion. Of course, this conversion can also be caused by the multiple reflexions of the wave in the complex geometry of the TORPEX vessel.

4.4 Visualization of the modes in the plasma

We have seen that the interaction of the plasma is more perceptible for the detection in the O-mode. Using the conditional averaged sampling method detailed in 3.3, we visualize in figure 4.13 and 4.14 the time evolution of CAS signal for each position of the RF detector. We can clearly see that we detect the evolution of the modes of the

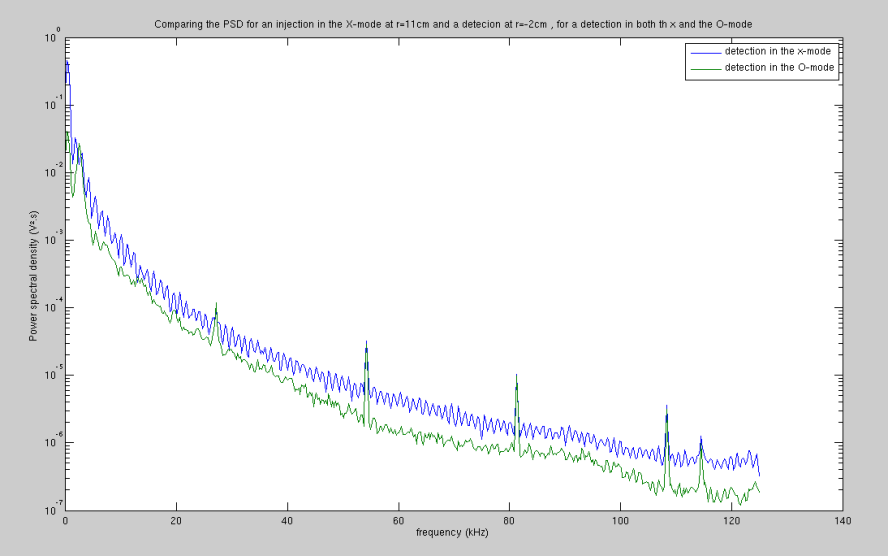


FIGURE 4.11: The spectral power density of the RF signal shows evidence of the interaction between the RF wave and the plasma modes around 2-6 kHz in the O-mode but not in the X-mode

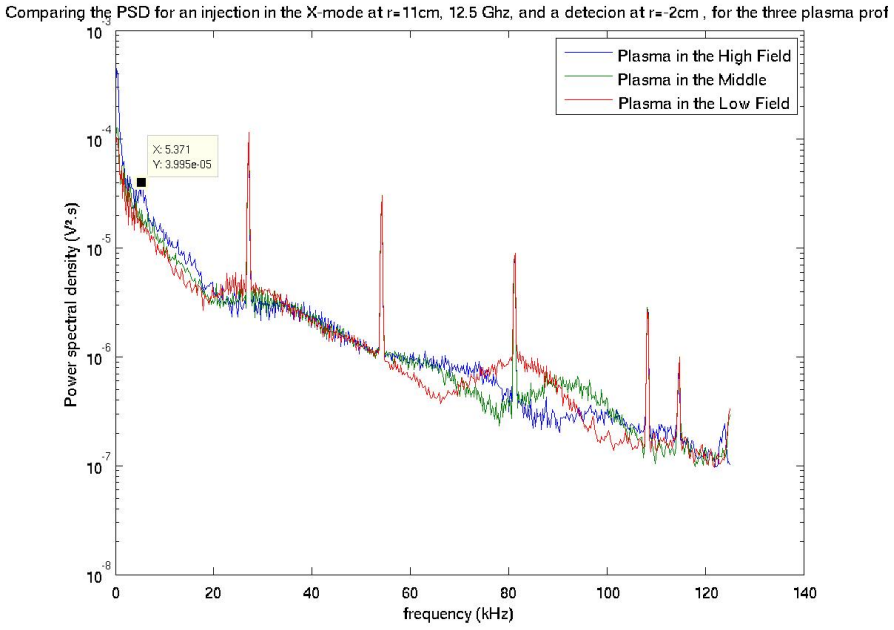


FIGURE 4.12: The spectral power density of the RF signal does not show any evidence of plasma modes in the X-mode at 12.5GHz

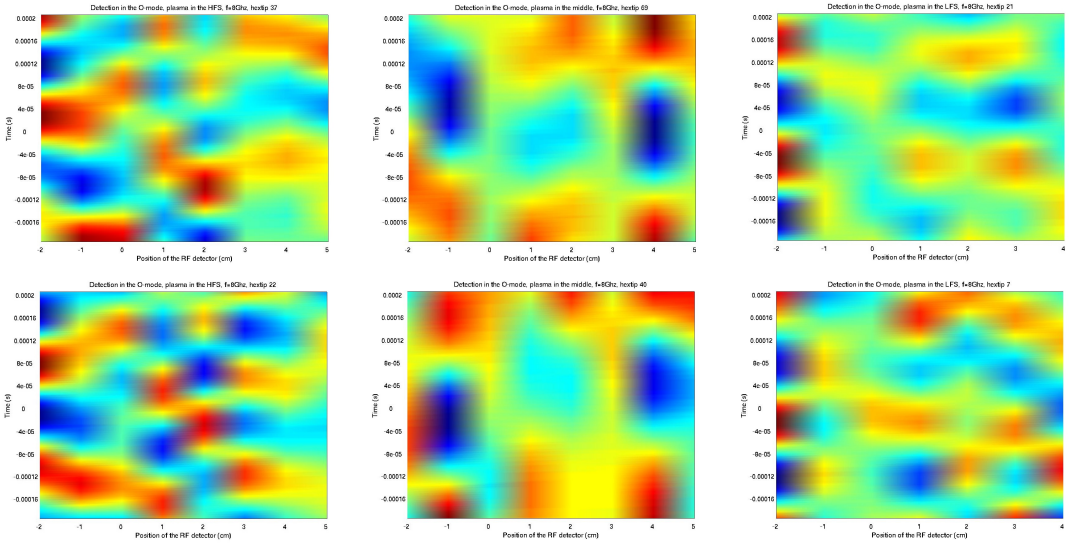


FIGURE 4.13: The different figures represent the time evolution of the CAS signal of the RF signal at different detection position using two neighbor probes (top and bottom results) on the same vertical line as a reference for an injection frequency of 8GHz. Both the RF and the HEXTIP signal were band-pass filtered before applying the CAS. From left to right: HFS, middle and LFS plasma

plasma at both 8 and 12.5 GHz. Moreover, the wavelength of the mode for the case of the plasma in the middle is twice as large as the other ones, which corresponds to the observations done in 4.3. Comparing the results obtained for two neighbor reference probes in the same vertical line, we see that the mode is vertically propagating. In fact, the time t=0s corresponds to a maximum in the amplitude of the plasma mode at the position of the HEXTIP probe used as a reference for the CAS. For instance, if we have a look at the left figures in 4.13, we can see that at the detection position r=-2cm, when the mode's amplitude is maximum for the upper probe (t=0s), the RF signal reaches a maximum whereas it is minimum when the mode's amplitude is maximum for the lower probe.

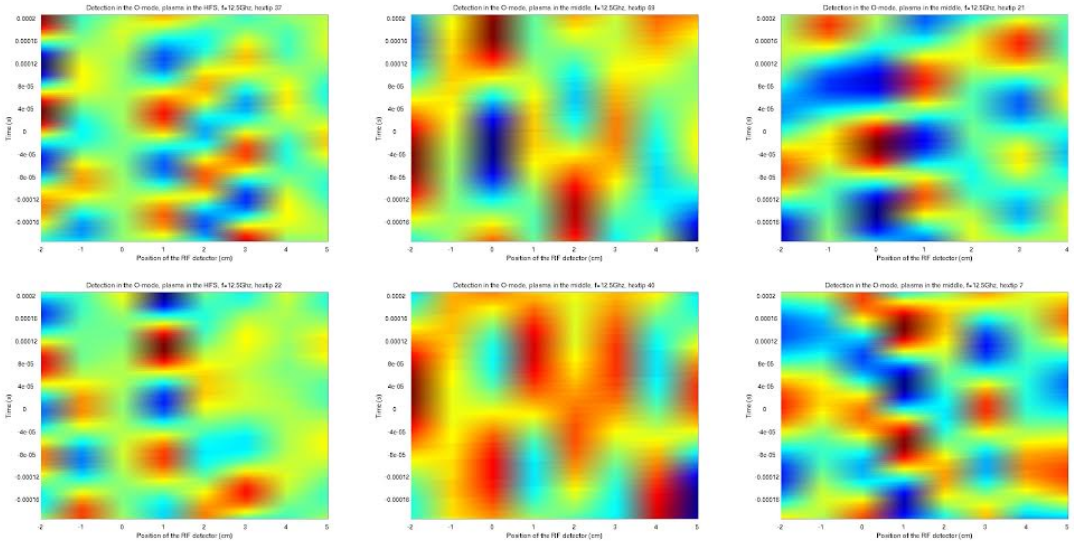


FIGURE 4.14: same as 4.13 with an injection frequency of 12.5GHz

Chapter 5

Conclusion

This work investigates the interaction between the RF waves in the X band and the TORPEX plasma. Using the commercial tool COMSOL, we have predicted a complex RF mode structure in the TORPEX cavity after the injection of the RF wave either in the O or in the X-mode. The behaviour of the TORPEX vessel being close to the one of a resonant cavity, this work can no longer be considered as a study of scattering.
The TORPEX experiment compaign has confirmed the existence of a mode structure during the injection of an RF wave in the TORPEX vessel. It has also revealed a strong influence of the plasma in on the RF signal, especially when the signal was detected in the O-mode. The plasma introduces large fluctuations on the RF signal which are characterized by a dynamic in the range of frequencies between 2 and 6kHz, depending on the plasma position.
Using the CAS analysis technique, we were able to visualize the temporal and spatial evolution of the plasma modes through the RF signal.
This work has opened a long to-do list:

in the experimental part:

- The strong interaction between the RF wave in the O-mode and the plasma modes suggests that the experiments should be done again with an injection in the O-mode.
- The detection system could be improved using a heterodyne-based detection system in order to have access to the electric field and the phase of the RF wave.
- One should try to increase the frequency and go more quasi-optical to study the scattering of the RF wave by the mode structures of the plasma. Here, however, one may encounter a problem due to the increasing transparency of the plasma with the frequency of an RF wave.

- Using a Vectorial Network Analyzer could bring more information on both the transmitted and reflected RF power with a dynamic range of 100dB.

in the simulation:

- Go further with the COMSOL simulation, try to reproduces different mode structures and analyze their influence on the detected signal to compare them with the experimental data.
- Take into account the tilt between the toroidal field and the TORPEX toroidal section in the dielectric permittivity tensor
- Make a 3D simulation that breaks the toroidal symetry of the 2-D simulation and take into account the real TORPEX geometry

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